Measuring the Task Value of Precision in Closed-Loop Agents

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Abstract

Precision upgrades in closed-loop agents can replace the current action, the subsequent controller, or both. We present an auditable same-state protocol that measures these interventions separately and compares higher-precision replacement with a same-precision alternative. On Qwen3-4B and Qwen3-8B in τ^2-bench, replacing a W4 agent's first tool-structure-divergent action with BF16 changes 15.5–20.9% of task–block outcomes, while net effects range from -1.9 to $+2.9$ percentage points. Crossed branches expose complementary upgrades and cases where continuation takeover undoes an action-level rescue. On ordinary BF16-sourced states of Mistral-Small-3.2-24B, turnover is approximately 10%, again comprising rescues and damages. Same-precision checkpoint substitutions also produce substantial turnover, while BF16's additional action value on common states remains unresolved. A frozen uncertainty trigger enriches outcome-changing decisions without demonstrating reliable net benefit. These findings separate intervention sensitivity, upgrade value, and the extra value of higher precision, providing a protocol for evaluating precision choices through paired task consequences.

1 Introduction

A quantized LLM agent is cheaper to serve than its BF16 reference, and a deployment can keep the reference as a fallback, as dynamic precision routing does (Li et al., 2026c). Suppose the quantized agent (Q) reaches a decision at which the BF16 agent (B) would act differently. There are three ways to spend B here: **replace only this step, let B take over from the next step, or both**. Average task success does not answer this. On our checkpoints, quantized-minus-BF16 success ranges from -0.2 to $+4.2$ pp across four W4 cells, and an equivalence test at ± 3 pp fails in every cell (Appendix B); even a precise end-to-end number would not say which step an upgrade should touch.

Figure 1a shows one such decision from our data. In an airline task, Q looks up a reservation id that does not exist and receives "not found"; B looks up the user's profile. We froze the state and ran all four combinations of {B's or Q's action} $\times$ {B or Q continuing}. Only B's action followed by B's continuation succeeds. When B supplies only this step, Q continues, searches for flights, writes a change and fails; when Q's failed lookup stands and B takes over, B fails too. Here the task value of B's action depends on the execution that follows it, and judging the action alone, however carefully, cannot reveal it.

More generally, the task benefit of a higher-precision upgrade can depend on which step is upgraded and on who executes afterwards, and whole-model swap evaluations mix these effects. Recent work already forks live agent trajectories and continues them with a different model, against same-model control forks (Gonuguntla, 2026); there, the new model writes the next action and everything after it, so the two cannot be told apart. Same-state crossed interventions measure them separately: we cross the source of the current action with the model that executes the rest of the episode, at a frozen state that replays bit-exactly, including the environment databases, the user simulator's view and the remaining step and error budget (§3). A same-precision substitution, a second W4 checkpoint of the same model in place of B, tests whether turnover and continuation dependence require a higher bit width, and benchmarks B's substitution against a fixed-precision alternative. Three questions thus need separate measurements: how sensitive the outcome is to a local replacement, how much net value a specific upgrade mode brings, and how much more a higher-precision replacement provides than a same-precision one. We study Qwen3-4B and

Qwen3-8B (Qwen Team, 2025) under simulated weight and KV-cache quantization in τ^2-bench (Yao et al., 2024; Barres et al., 2025), BFCL multi-turn (Patil et al., 2025) and ALFWorld (Shridhar et al., 2021), and Mistral-Small-3.2-24B (Mistral AI, 2025) on ordinary BF16-sourced τ^2-bench states.

Findings.

- **Local replacement is sensitive; its net value is small** (§4, §7). Replacing a W4 agent's first tool-structure-divergent action with B's changes the outcome in 15.5–20.9% of task×block units, with rescues and damages of similar size and a net effect of -1.9 to $+2.9$ pp (Figure 1b). At these states (exploratory, post hoc), changing the action changes the outcome under some continuation in 34.9% of states, and changing the continuation controller does so under some action in 29.6%; the two sets overlap on the 26.9% where the swap's effect depends on the continuation. Only the joint upgrade succeeds in 4.0%, and B's takeover undoes a rescue by B's action in 4.8%. Paired comparisons separate the upgrade modes only on the fixed task pool and only in some cells, not in the W4 pool (exploratory; Figure 2). On ordinary BF16-sourced states of Mistral-Small-3.2-24B, replacing the action turns over about 10% of states, in both directions, with no resolved net value.
- **The extra value of higher precision is not established** (§5). Swapping only the calibration checkpoint, at W4 and at W3, also produces two-sided flips and continuation dependence at comparable size, and BF16's additional action value over a second W4 checkpoint on common states is not resolved. The two substitutions share more affected units than the tasks' flip propensity explains, yet at W4, 138 of the 245 affected units respond to only one of them. Equivalence of the two turnovers within ± 3 pp is not shown.
- **Detecting sensitivity is not selecting value** (§6). A frozen Q-only uncertainty trigger captures more outcome-changing decisions than random selection, but rescues and damages alike: 2–5 of the 34–38 net units that a hindsight oracle with the same budget shows are available.

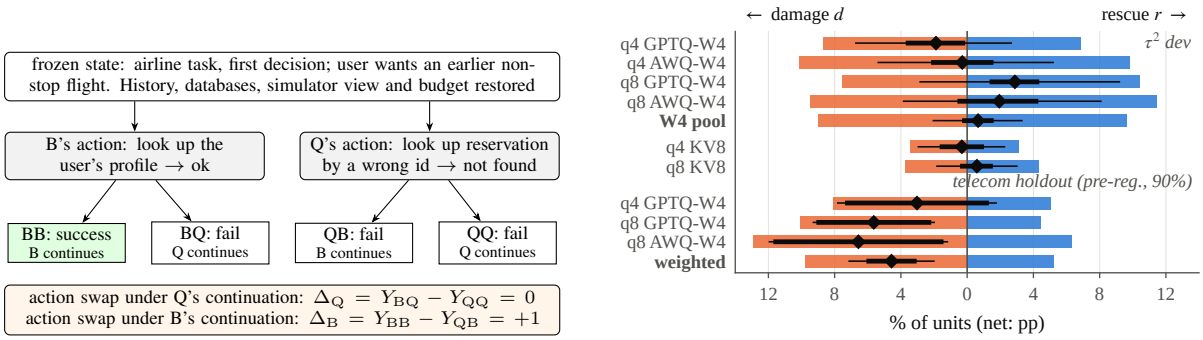
Contributions. (1) *An executable same-state intervention protocol* that measures separately the value of upgrading the current action, the continuation, and both, with a same-precision reference substitution. The factorial design is standard; the contribution is making these counterfactuals executable and auditable in closed-loop agents (exact state restoration, budget accounting, replay acceptance, explicit denominators). (2) *Evidence that the three questions have different answers*, summarised in the findings above.

2 Related Work

Quantized agents compared episode by episode. *Flat Score, Amplified Failures* (Jang et al., 2026) is the closest study in setting: on τ^2-bench retail and telecom, 4-bit agents keep a flat task score while process damage grows, with a two-sided paired task split and recovery equally frequent at both precisions (Appendix P). Its episodes are sampled independently, so the precision of the action that went wrong and that of the controller that must recover change together; we hold the state fixed and vary the two separately. Two-sided flips under compression were first shown on static tasks (Dutta et al., 2024); agent-compression benchmarks report averages (Dong et al., 2025).

Forks and counterfactual replay. *The Replay Gap* (Gonuguntla, 2026) evaluates per-step model switching for routers: it forks live SWE-bench Verified trajectories, continues each fork with another model, and compares against same-model control forks (Appendix P). Live forking and a same-model control are therefore not new. Its swap replaces the next action and the continuation together, which is our "both" contrast, and its control holds the model fixed to measure sampling and replay noise. We cross the action source with the continuation controller at a bit-exact frozen state, and our reference changes the model while holding the bit width fixed. Restoring a state, replacing one action and letting the original policy continue locates decisive errors (Zhang et al., 2025), attributes failures through a do-intervention on one step (Shah, 2026), and turns alternative actions into training signal (Li et al., 2026a,b). This is our "this step only" contrast, and the factorial design itself is standard. We found no work among the 32 closest that crosses the action with the continuation at a precision-induced divergence or reads the result against a same-precision alternative.

Precision routing and its signals. DMR (Li et al., 2026c) learns a per-step BF16/W4 router



(a) (b)

Figure 1: Four-branch task value. (a) A real decision (Qwen3-4B, AWQ-W4; airline task 31, decision 1 of 12): B's action is worth a success only if B also continues. (b) Replacing the first tool-structure-divergent action with B's, per cell: rescues r (right) and damages d (left) in % of all task×block units, and the net $G_{\mathrm{all}} = r - d$ (diamond, pp). Thick/thin: fixed-pool/new-task 95% intervals on the τ^2-bench development panel; block-t/crossed 90% on the pre-registered telecom holdout ("weighted" is its decisional endpoint). All cells: Appendix Table 2.

from BF16-versus-quantized KL supervision and episode reward. Sensitivity has also been localised to phases (Lu et al., 2026) and task types (Zhang et al., 2026), and for quantized reasoning models to high-entropy, high-KL token positions (Lotfi et al., 2026). We do not propose a router. We measure the task value a router should learn, and our simple frozen trigger (§6) does not test, let alone refute, a learned router such as DMR.

Evaluation noise and user simulators. Prefix caching changed multi-turn tool-use trajectories in 36.2% of episodes at 16-bit and 75.0% at 4-bit (Patodiya, 2026); agentic evaluations vary from run to run (Bjarnason et al., 2026; Chen et al., 2026) and with the user model (Seshadri et al., 2026; Jain and Sandhu, 2026).

3 Measuring the Task Value of a Replacement

3.1 Setting

Tasks and agents. We use τ^2-bench in text mode: a development panel of 58 retail and 27 airline tasks plus 80 telecom development tasks, and a holdout of 34 telecom tasks that no agent ran before its protocol was frozen. The agents are Qwen3-4B-Instruct (q4) and Qwen3-8B with thinking disabled (q8); §7 adds Mistral-Small-3.2-24B. One greedy Qwen3-30B-A3B-Instruct user simulator serves all agents. Retail and airline success is the final-database match; telecom uses the official task-basis success. BFCL multi-turn and ALFWorld, which have no user simulator, run once per item (Appendix A).

Quantization and serving. Q is a simulated (fake-quantized) checkpoint served with BF16 compute on the same path as B: GPTQ (Frantar et al., 2023) and AWQ (Lin et al., 2024) at W4 and a KV8 cache in the main cells; GPTQ-W3, RTN-W3 and a rotated KV4 cache as boundary (stress) cells. B and Q run on vLLM (Kwon et al., 2023) with batch-invariant kernels (He and Thinking Machines Lab, 2025) and deterministic compilation in separate server processes, so no prefix cache is shared. A *block* is one closed-loop run of Q over the task pool under one prompt nonce (a random session-identifier line in the system prompt); blocks are the replicate unit.

3.2 Four branches on one state

Frozen state. Before Q's decision t we restore the state s_t: the full message history; the environment and user-device databases, rebuilt by strict replay of every recorded tool call; the user simulator's visible conversation; and the used step and error budget. B and Q generate their actions from identical prompt token ids, and the state is written to disk before any branch runs.

Crossed intervention. $Y_{\mathrm{XY}} \in \{0, 1\}$ is task success when the action at s_t comes from X and model Y executes the rest of the episode, X, Y $\in \{\mathrm{B}, \mathrm{Q}\}$ (Figure 1a). The action is executed by the real environment or answered by the real simulator, and all four branches start with the same remaining budget. Each upgrade option is a paired

contrast with QQ:

$$G = Y_{\mathrm{BQ}} - Y_{\mathrm{QQ}} \quad \text{(this step only)},$$
$$H = Y_{\mathrm{QB}} - Y_{\mathrm{QQ}} \quad \text{(from the next step)},$$
$$F = Y_{\mathrm{BB}} - Y_{\mathrm{QQ}} \quad \text{(both)}.$$

The paired effect of the action swap under each continuation is $\Delta_{\mathrm{Q}} = Y_{\mathrm{BQ}} - Y_{\mathrm{QQ}}$ $(= G)$ and $\Delta_{\mathrm{B}} = Y_{\mathrm{BB}} - Y_{\mathrm{QB}}$; their difference $I = \Delta_{\mathrm{Q}} - \Delta_{\mathrm{B}}$ is the action–continuation interaction. Δ_{Q} and Δ_{B} take values in $\{-1, 0, +1\}$, so $I \in \{-2, \ldots, +2\}$. When $\Delta_{\mathrm{Q}} \neq \Delta_{\mathrm{B}}$ ("off-diagonal"), the swap's value depends on the continuation; a *strict reversal* is $\Delta_{\mathrm{Q}} = -\Delta_{\mathrm{B}} \neq 0$. For the action swap, a *rescue* is $(Y_{\mathrm{BQ}}, Y_{\mathrm{QQ}}) = (1, 0)$ and a *damage* is $(0, 1)$; with rates r and d, the turnover is $M = r + d$ and the net gain $r - d$. All contrasts measure effects on task success, not which action is correct.

Same-precision alternative. A large M or a frequent interaction could mean that this decision needs higher precision, or only that it is sensitive to any change of model. To tell these apart, we rerun the design with Q′, a second GPTQ-W4 checkpoint of the same model calibrated on a different sample, in place of B: the same Q episodes, tasks and nonce blocks, and a protocol frozen before any outcome (q4 and q8 GPTQ-W4; GPTQ-W3 as a stress check). Turnover M', rescues r', damages d' and the off-diagonal share are defined as above with Q′ for B.

3.3 Making the counterfactuals auditable

Replay acceptance. A causal reading requires exact replay. Before any development outcome was analysed, an engineering gate on real divergence states passed every check: Q→Q no-ops matched in 20/20 states and B→B no-ops in 10/10, in every prompt-token hash, message, database hash, termination, reward and budget counter. In production, every fork task re-ran the QQ no-op on an outcome-independent 10% hash sample of its states, and one mismatch would have quarantined the task's whole output; all 924 sampled development states passed, as did all sampled states of the other banks and of the same-precision run (Appendix A). Exactness is a tested property of this serving stack, not a general one (Yuan et al., 2025): greedy decoding on vLLM 0.29.0 with batch-invariant kernels and deterministic compilation, and simulated checkpoints on dense BF16 kernels; untested conditions and the determinism tests are in Appendix A.

Which states. *Bank P*: for each task, block and cell, the first decision at which a_{B} and a_{Q} differ in kind (tool call, text, stop), tool name, call count or order, or normalised arguments; differences in text only are skipped. *Bank U*: one ordinary decision per task, drawn by a hash fixed before generation, in three nonce blocks (U2B: up to six in §5 and §7), from Q's own trajectories (U1) or from trajectories shared by all cells of a model (U2B: BF16; U2Q: Q).

Denominators and inference. We never mix three denominators. (1) *Full pool*: every task×block unit, units without a Bank P state counting as zero; r, d, M and $G_{\mathrm{all}} = r - d$ use it. (2) *Divergent states*: the Bank P states of a cell; G, H, F and the off-diagonal share use it. (3) *Bank U*: one state per task. Within a cell all branches share each state and all tasks of a block share its nonce, so within-cell contrasts are paired and identified; cross-cell comparisons are descriptive, because each cell selects its own divergences. The primary interval treats the task pool as fixed (block bootstrap with at least 10 complete blocks, else t over block means); a crossed task×block bootstrap is the interval for new tasks. Only complete blocks enter.

4 Precision Replacement Redistributes Success and Failure

Many rescues, many damages, small net. On the τ^2-bench development panel (85 tasks × 20 complete blocks), replacing a W4 agent's first tool-structure-divergent action with B's changes the task outcome in $M = 15.5$–20.9% of units (Figure 1b). Rescues and damages are of similar size, $r = 6.8$–11.4 and $d = 7.5$–10.1 pp, so the net gain G_{all} lies between -1.9 and $+2.9$ pp. Two fixed-pool intervals exclude zero (negative for q4 GPTQ-W4, positive for q8 GPTQ-W4); no new-task interval does. Pooled over the four W4 cells, $G_{\mathrm{all}} = +0.7$ $[-0.3, +1.6]$ pp with $M = 18.6\%$. KV8 cells turn over fewer outcomes ($M = 6.5$–8.1%, a descriptive comparison). G_{all} is a local replacement effect, not an end-to-end comparison (Appendix B).

Which intervention changes the outcome. On divergent states, the average estimates of the action swap are the same under Q's and under B's continuation (W4 pool: $\Delta_{\mathrm{Q}} = G = +0.5$ $[-0.7, +1.7]$, $\Delta_{\mathrm{B}} = +0.5$ $[-1.0, +2.1]$ pp), and no

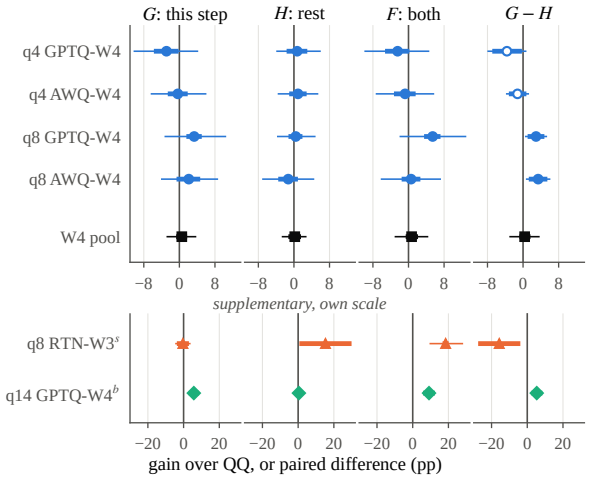


Figure 2: Upgrade modes on the same divergent states: replace this step (G), take over from the next step (H), or both (F), as gain in success over QQ, and the paired difference $G - H$; pp. Top: the four core W4 cells and their pool (τ^2-bench development); unadjusted, thick: fixed-pool 95%, thin: new-task 95%, except for the cells' $G - H$ (the Holm family), where thin is the simultaneous interval and only filled markers are significant after Holm adjustment. Bottom, own scale: configurations outside the family ([s]stress cell, three blocks; [b]BFCL, one run per item). Interval details: Appendix D.

average gain is detected for upgrading this step, the rest, or both ($H = +0.2$, $F = +0.7$ pp, all intervals covering zero). State by state, the four outcomes show which intervention matters (exploratory, post hoc; W4 pool; % of divergent states; fixed-pool, then new-task 95% intervals). Changing the action changes the outcome under at least one continuation in 34.9% ([33.6, 36.0]; [28.8, 41.0]), and changing the continuation controller does so under at least one action in 29.6% ([28.5, 30.7]; [24.2, 35.3]). The two sets overlap exactly in the 26.9% of states where the swap has a different effect under the two continuations ([25.7, 28.1]; [22.1, 32.0]), so the shares must not be added; all four arms agree in 62.4%. In 4.0% ([3.5, 4.5]; [2.7, 5.3]) only the joint upgrade succeeds (joint rescue, as in Figure 1a), and in 4.8% ([4.1, 5.4]; [3.2, 6.5]) B's action rescues under Q's continuation while B's takeover of the continuation undoes the rescue (rescue undone). These shares describe operational consequences of specific upgrade modes, not by themselves a distinct interaction mechanism (five classes, a label-permutation reference, task share of variance and tests: Appendix F).

Paired comparisons distinguish upgrade modes. All three upgrade modes are measured on the same states, so they can be compared by paired differences (Figure 2). Over the four core W4 cells, $G - H$ is a fixed Holm family ($m = 4$; fixed task pool, t over blocks). In both Qwen3-8B cells, action replacement outperforms continuation takeover: $G - H = +2.9$ (GPTQ-W4) and $+3.4$ pp (AWQ-W4), Holm-adjusted $p = .013$ and $.012$; this ranks the two modes and does not show that replacing the action beats QQ. In the Qwen3-4B cells the point estimates favour continuation takeover (-3.6 and -1.3 pp), but neither is significant after Holm adjustment. All four new-task intervals include zero, so the finding is limited to the fixed task pool and stays exploratory (intervals and further cautions: Appendix D). In the low-success stress cell q8 RTN-W3, continuation takeover beats action replacement ($G - H = -15.7$ [-27.5, -3.9] pp, three blocks). For Qwen3-14B GPTQ-W4 on BFCL (one exploratory configuration), action replacement wins ($G - H = +5.3$ [$+2.3$, $+8.4$] pp), and action and continuation are complementary there (Appendix D). The joint upgrade F has the largest point estimate in both examples, and we compare no compute costs, so no cost-optimal mode is established.

Other states, tasks and environments. On BFCL multi-turn and ALFWorld, which have no simulated user (one run per item), the replacement turns over 5.2–24.6% of items in both directions, and the off-diagonal share is 4.7–17.1% in BFCL W4 cells and 32.8–40.0% in q8 ALFWorld W4 cells (post hoc; Appendix G.3). On 34 unseen telecom tasks, under a protocol frozen before any holdout unit ran (12 fresh blocks, three W4 cells), the replacement again rescued and damaged ($r = +5.21$, $d = +9.77$ pp), and task fragility repeated across nonce halves ($\rho = 0.59$). The primary $G_{\text{all}} = -4.57$ pp (crossed 90% [-7.17, -1.96]) is *inconclusive* by the pre-registered readings, which had no negative category; outside them, the interval lies below zero (Appendix N). At ordinary (Bank U) states, turnover is lower (W4 cells, U1 and U2B: 4.7–9.0%), the net gain covers zero, and most outcome changes come from differences in text, whose labels did not reproduce under a second user simulator, so we do not report them as a property of the action (Appendix J).

5 Does an Upgrade Need a Higher Bit Width?

A large turnover or a frequent interaction is tempting evidence that a state needs higher precision. If these consequences required a higher bit width, the same-precision substitution (§3.2) should produce markedly fewer of them. Because each design forks at its own first divergence, the two may select different states, so shared and exclusive flips cannot be split into effects caused and not caused by precision.

A higher bit width is not necessary for two-sided flips and continuation dependence. They also appear when only the calibration checkpoint is swapped (Figure 3a). At W4 (frozen final analysis set, $K = 6$ blocks), the second checkpoint's action rescues $r' = 9.3$ [7.0, 11.7] and damages $d' = 8.4$ [6.4, 10.5] pp of units ($M' = 17.7\%$ against $M = 16.8\%$ for B), and 26.2% of its divergent states are off the diagonal, against 25.0% for B; each same-precision rate has a 95% lower bound above zero. In the W3 stress pair the turnovers are 17.3 and 18.8 pp and the off-diagonal shares 27.6% and 27.3%. Neither pair shows the two turnovers to be equivalent within ± 3 pp (Appendix D.6), so flip rate and interaction size alone cannot diagnose a need for a higher bit width.

Units affected by both, or by only one, substitution. At W4 (Figure 3b), 107 units flip under both substitutions, 64 only under B and 74 only under the second checkpoint. The reference that matters keeps each task's flip propensity: a within-task block shuffle gives 76 [69, 84] units flipping under both, so the substitutions share affected units beyond the fact that the same tasks are flip-prone anyway. The shared core is on the action side: at W4, B's and the second checkpoint's rescue sets agree with $\kappa = 0.60$ and their harm sets with $\kappa = 0.54$. When B also takes over the continuation, the affected units are largely substitution-specific (Appendix D.6). The difference in net value comes only from the substitution-specific units and is not resolved (-1.6 [-6.2, $+3.0$] pp). These analyses are descriptive (Appendix D.6, Figure 8).

Common states: sensitivity, extra value and gain are different quantities. A pre-registered comparison holds the states fixed: on 510 ordinary q8 states (85 tasks $\times$ 6 blocks), the current action comes from B, from Q or from a second W4 checkpoint Q′, and B or Q continues (Appendix Figure 7). Under Q's continuation, replacing Q's action by B's flips 7.1% of outcomes for a net $+1.2$ pp; replacing it by Q′'s flips 8.0% for a net $+0.2$ pp. What B's action adds beyond another W4 action, $K(c) = V(\mathrm{B}, c) - V(\mathrm{Q}', c)$, is $+1.0$ pp under Q's continuation and $+0.8$ pp under B's. After the fixed extension from three to six blocks, neither an advantage nor equivalence within ± 3 pp is established (95% t intervals [-3.5, $+5.4$] and [-1.8, $+3.3$]; a two-look sensitivity analysis widens them), and a separate q4 analysis on two blocks is also not resolved (Appendix D.6). A second W4 action thus changes about as many outcomes as B's, with a net effect within ± 3 pp of zero, so how sensitive the loop is to an action, what higher precision adds at a state and what a given upgrade gains have to be measured separately.

6 Does Sensitivity Convert into Upgrade Value?

G, H and F require running B and all four branches; a deployable rule sees only Q. We test whether a Q-only signal that finds outcome-changing decisions also finds the ones worth upgrading.

Uncertainty locates divergence and outcome changes. Scored at the token level on one development block, Q's entropy at the start of its action separates states where B and Q act differently from states where they act identically (AUC 0.66–0.91 per sub-bank and model), but among states where the actions differ, no Q-visible statistic reliably separates flips from non-flips (Appendix D.5). We then froze a Q-only trigger on that block before any later state was scored: escalate one decision when the minimum Q margin over its action is low (margins predict cross-precision divergence; Du et al., 2026), with per-cell thresholds at 5, 10 and 20% escalation rates. On two held-out blocks (1020 rows, 69 outcome-changing states) it captured 8, 12 and 27 outcome-changing states, against 3.48, 7.52 and 14.30 expected at random ($p = .0182, .0613, .0002$; Figure 4b).

It does not yet select net-beneficial replacements. The captured states were rescues and damages alike (6/2, 7/5 and 16/11), for net gains of $+4$, $+2$ and $+5$ against random expectations of $+0.21$, $+0.98$ and $+2.38$ (exact one-sided $p = .0345, .417, .262$); no level was fixed in advance,

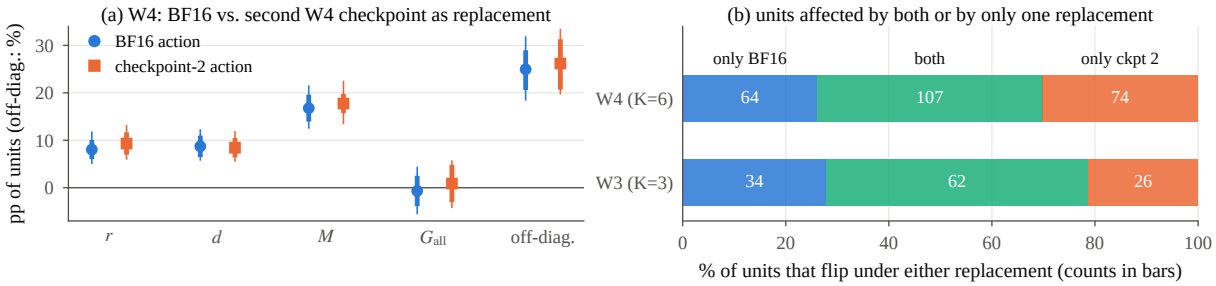


Figure 3: Same-precision reference substitution. (a) W4 (q4 and q8 GPTQ-W4 pooled; 1020 task×block units; $K = 6$ blocks): replacing Q's first divergent action with B's (circles) or with a second W4 checkpoint's (squares); rescues r, damages d, turnover M and net G_{all} in pp of units, off-diagonal share in % of divergent states. Thick: fixed-pool 95% (t over block means); thin: new-task 95% (crossed). (b) Units whose outcome flips under either substitution's action replacement (G): only under B's, under both, or only under the same-precision checkpoint's; W4 and the W3 stress pair (510 units, three blocks). Equivalence tests, per-model numbers: Appendix D.6.

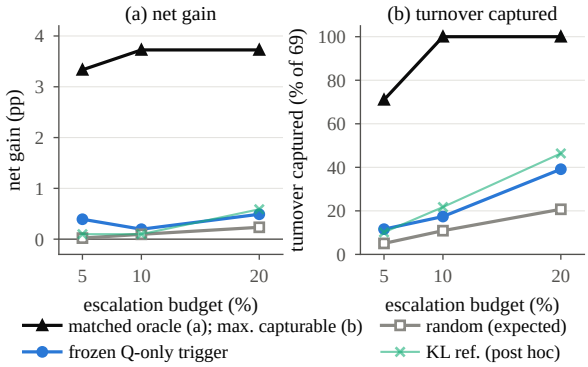


Figure 4: Sensitivity versus benefit, by escalation budget, on held-out ordinary states (1020 U1 rows; 38 rescues, 31 damages): a frozen Q-only uncertainty trigger, matched-count random selection (expected), a hindsight oracle with the same per-cell counts, and a faint post-hoc KL(B‖Q) reference (needs BF16). (a) Net gain, (rescues − damages)/1020, pp. (b) Share of the 69 outcome-changing states selected; black: the most any selection with the same per-cell counts can capture. Values: Table 13.

and the 5% rate does not survive Holm correction. A hindsight oracle with the same per-cell counts nets +3.33, +3.73 and +3.73 pp (Figure 4a). This headroom is positive by construction, because the oracle picks only rescues and never a damage; the trigger captures 4/34, 2/38 and 5/38 of it. A post-hoc KL(B‖Q) score, which needs BF16, also captures turnover above chance without a significant net gain (Appendix L). The held-out blocks reuse the development tasks. A learned router trained on task reward (Li et al., 2026c) is not tested here.

7 When Upgrade Value Appears, Disappears or Moves

Upgrade value can appear, disappear, or shift between the action and the continuation. In §4, without a simulated user (BFCL, ALFWorld) and on the pre-registered telecom holdout, the replacement rescues and damages and its net value is small or negative; in one exploratory configuration (Qwen3-14B, BFCL) value can appear in the action, and in a three-block stress cell (q8 RTN-W3) it can move to the continuation. Stress cells are kept separate from W4 (Appendix H).

Three further lines vary numerical quality, cache location and model family. They use ordinary BF16-sourced states (Bank U U2B, §3), drawn outcome-blind from the BF16 model's own trajectories and shared by every method of that model; they follow the BF16 policy's state distribution, not the one a quantized agent reaches. Methods are compared on these states by $D_{cc} = Y_{\text{QB}} - Y_{\text{BB}}$, Q's action against B's under the shared BF16 continuation (details: Appendix I).

Numerical quality: better local numerics, no detected task value. Of two rotation-based quantizers (Ashkboos et al., 2024; Liu et al., 2025), SpinQuant-W4A4 is closer to BF16 than QuaRot-W4A4 on the shared q8 states (mean token-level KL 0.165 against 0.199), but its task value is not shown to differ: the pre-registered matched contrast is +1.2 [−3.9, +6.2] pp (Holm-adjusted $p = 1.0$; one block rests on a re-run). A decomposition on the same 255 states (exploratory, post hoc; t over three block differences) resolves the KL gap (-34 [-57, -11]$\times 10^{-3}$) but neither how often Q's action differs from B's (+0.8 [−11.4,

+13.0] pp) nor how often a difference flips the outcome (+2.7 [−10.2, +15.3] pp). Across the eleven q8 cells (Spearman, descriptive), mean KL orders the action-difference rate ($\rho = 0.99$) and, through it, B's rescues and damages together (0.81 and 0.93), but not the flip rate given a difference (0.46) or the net D_{cc} (−0.44); within cells it separates differing from identical actions (mean AUC 0.93), not flips among differing actions (0.46). Local fidelity thus predicts how often an upgrade changes the action, not whether the change matters or in which direction (Appendix I). B's action alone (G) is resolved in no cell. Fixed-pool intervals for B's takeover H or full substitution F exclude zero in weight-only (AWQ-W4, QuaRot-W4), W4A4 and KV8 cells, and new-task intervals still do in both W4A4 cells and KV8 (unadjusted), so resolved value is not confined to quantized activations. It is largest in SpinQuant-W4A4 ($H = +11.8$ [6.7, 16.8], $F = +13.7$ [9.3, 18.2] pp).

Cache location: no detected shift of value. KIVI (Liu et al., 2024) quantizes keys per channel and values per token; we use a replay-compatible variant. KIVI-2 against KV8 is not resolved (−0.6 [−38.0, +36.8] pp), nor is KIVI-4 against KV8 (+1.6 [−2.9, +6.0] pp). The KIVI-2 contrast rests on two blocks, both depending on re-runs of arms lost to server failures (strict rule: not estimable), and the unquantized kernel path itself reproduces B's action in 85.1% of states.

Model family: two-sided turnover on Mistral, no resolved net value. On Mistral-Small-3.2-24B, at its own ordinary BF16-sourced states, replacing Q's action with B's again flips outcomes in both directions with no resolved net value. The estimates are conditional on the 83 tasks with a complete, recoverable BF16 source state in every block, fixed by a dated protocol amendment (GPTQ-W4 on six blocks, RTN-W4 on five). In GPTQ-W4, B's action rescues $r = 4.6$ [2.4, 6.8] and damages $d = 5.4$ [2.4, 8.5] pp of states ($M = 10.0$ [7.0, 13.1]; fixed-pool 95%), and no mode's net value is resolved ($G = -0.8$ [−5.2, +3.6], $H = +1.4$ [−3.0, +5.8], $F = +0.0$ [−4.2, +4.2]). RTN-W4 is similar ($M = 10.4$ [6.3, 14.4]). The pre-registered contrast GPTQ − RTN is not resolved: $\Delta D_{cc} = +2.2$ [−3.5, +7.8] pp, $p = .35$, on five blocks, two of which are admitted only by a scoring rule adopted after an interim look (Appendix I).

8 Discussion

Sensitivity, upgrade value and the extra value of precision are separate questions. How often a local replacement changes the outcome, how much net value a specific upgrade mode brings, and how much more a higher-precision replacement provides than a same-precision one need separate measurements. Replacing one action changes outcomes in both directions: many, for little net value, at Qwen3's first tool-structure divergences, and about 10%, with no resolved net value, at Mistral's ordinary BF16-sourced states (§4, §7). The action and the continuation both change outcomes, on overlapping sets of states, and on the fixed task pool paired comparisons separate the upgrade modes in some cells but not in the W4 pool. These are operational consequences of specific upgrade modes, not by themselves evidence of a distinct interaction mechanism (Appendix F).

Changing outcomes is not improving them. A second checkpoint at the same bit width, at W4 and at W3, changes outcomes about as often as B does, on a shared core of units plus many that only one substitution affects (§5); a flip shows only that the outcome at that state is sensitive to the action, not that it needs more bits. On identical states, B's action is shown neither worth more than the second checkpoint's nor equivalent to it, and a signal that finds flips does not yet find net gains (§6). We do not conclude that precision is irrelevant: neither reference shows equivalent turnovers, and B's single-cell values stand.

Recommendations for evaluating precision interventions in agents. (i) Report the rescues and damages of the specific upgrade mode (the action, the continuation, or both) on the same states, not only success rates or flip rates. (ii) Include a same-precision reference substitution, so that sensitivity is not read as a need for precision. (iii) Give intervals for both the fixed task pool and new tasks (crossed), because the conclusions can differ, as for the W4 $G - H$ family (§4).

Limitations

What we do not claim. We do not claim success equivalence between BF16 and W4 agents; equivalence tests at ± 3 pp on new-task intervals fail. G, H and F are oracle quantities, not the value of a deployable policy (§6). We do not claim that BF16 and a second W4 checkpoint are equivalent replacements. We do not claim a W$\times$KV interaction, and we do not present static precision restoration as a method (Appendix O). We do not claim a distinct action–continuation interaction mechanism, or that outcomes are mainly determined by the action: the outcome classes and the label-permutation reference (Appendix F) are exploratory and post hoc, with headline shares from the retail and airline development panel, and the action-over-continuation gap comes only from the 13.8% of W4 divergent states with exactly two successful arms.

Two model families, limited coverage. Most crossed forks use Qwen3 models: 4B and 8B throughout, plus a single Qwen3-14B configuration on BFCL. The second family, Mistral-Small-3.2-24B (GPTQ-W4, RTN-W4), was forked only at ordinary BF16-sourced τ^2-bench states of an 83-task conditional population (§7); it has no first tool-structure-divergence forks, no same-precision reference and no outcome-class analysis. Effects may differ at other scales and in other families. End-to-end runs also exist for Qwen3-14B and Qwen3-32B; apart from that one Qwen3-14B configuration they have no crossed forks, and we draw no conclusion from them (Appendix H). A time-boxed Llama-3.1-8B-Instruct check (GPTQ-W4, BFCL multi-turn) was stopped at its gate: BF16 solved 5 of 120 gate items, so the four arms carried almost no outcome variance, and it yields no evidence for that family.

Simulated quantization. Quantized checkpoints are dequantized and served with BF16 compute on a deterministic path. Deployed low-bit kernels change the numerics and add serving variation that we deliberately remove. We make no claims about latency, memory or deployed kernels, and the cost of a BF16 fallback is outside our scope.

One user simulator; two small controls. All τ^2-bench episodes use a single greedy simulator, and continuation effects include its replies. Both second-simulator controls replace only the user after a fixed history, cover 78 and 80 states, and say nothing about real users. The Bank P control has 20 states per cell, too few to confirm or refute few-pp contrasts. Neither control can be reproduced draw for draw. The replacement user is a hosted Sonnet 5.5 model sampled with its default settings, without temperature or seed control. In the Bank U control, Claude Code subagents answered many relay requests in one context, so a responder could see other arms or replicates of the same conversation. The Bank P control removed this exposure by issuing one isolated, single-request call per user turn. The same-precision reference (§5) covers q4 and q8 GPTQ-W4 on the development panel (six blocks: the frozen rule takes the longest complete common prefix, and block f07 of the same-precision q8 design is incomplete) and GPTQ-W3 on three blocks, and swaps one calibration sample for another; it is our only reference for how much turnover and continuation dependence a change of model without a change of bit width produces, and it says nothing about AWQ, KV8, RTN or rotated KV4.

LLM annotation. Process labels in Appendix M come from LLM annotators (Sonnet 5.5, two independent instances plus an adjudicator) and rule-based checks. No human annotation has been done. Agreement among instances of one model family shows reproducibility, not correctness.

Small confirmation and partial banks. The pre-registered holdout covers 34 telecom tasks and three cells. Its pre-registered readings had no negative category, so a clearly negative estimate could only be read as inconclusive; its negative point estimate, inconclusive by those readings, applies only to telecom, these three cells and this simulator (its replay acceptance was audited post hoc and is complete; Appendix N). Most development results come from retail and airline, so the holdout confounds a domain shift (telecom) with generalisation to new tasks. On 80 telecom development tasks, the two q8 holdout cells show no detectable net effect (Appendix D.3), so the telecom domain alone does not produce a detectable drop; but their 6-block intervals are wide and q4 has no telecom development forks, so the holdout drop may reflect heterogeneity among its 34 tasks as much as a domain effect. Bank P intervenes only at the first tool-structure divergence per task and block. Bank

U samples one state per task and block; its main analyses use three blocks, so their block intervals rest on two degrees of freedom, and the later panels use two to six blocks. Prompt nonces define replicate blocks; they are a researcher-chosen perturbation, not a deployment distribution.

Ethics Statement

This work involves no crowdworkers and no personal data. All users are simulated by an LLM on public benchmark tasks, and all agents are openly released Qwen3, Mistral-Small-3.2 and (gate check only) Llama-3.1 checkpoints, or simulated-quantized versions of them. The process annotations were produced by LLMs on synthetic benchmark conversations. The main risk we see is over-reading. A small forked-state gain on one simulator and a fixed task pool does not certify a quantized agent for deployment, and our end-to-end comparisons do not establish equivalence. Deployment decisions in consequential settings should not rest on these measurements alone.

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A Setup Details

Splits. The development split has 58 retail, 27 airline and 80 telecom tasks. The holdout has 34 telecom tasks, and its pre-registration hash is recorded in the split file. The telecom split was fixed from τ^2-bench's 114-task base split by a stratified, salted hash before any run. The external benchmarks use BFCL multi-turn (base, miss_func, miss_param and long_context, 197 items each) and ALFWorld (134 out-of-distribution games). Items were selected by hash in manifests committed before any model run.

Success endpoints. For retail and airline we use the final-database match with normal termination. This is not τ^2-bench's official reward, which additionally scores natural-language or communicate checks. Telecom uses the official deterministic task-basis success. Context overflow and unparseable output count as failures. Infrastructure outages are missing, never failures, and a block with a missing unit leaves the primary analysis.

Fork mechanics. The forced action is returned by a fork agent for the one step that follows h_t. A tool call is executed by the real environment, and text is answered by the real simulator, which never reuses observations from the original trajectory. τ^2-bench resets step and error counters when it starts from a history. The fork restores them to the values used at t, so the remaining budget equals the original. Every max_steps and too_many_errors episode on the checked blocks is reproduced by the recount.

Replay acceptance details. In the engineering gate, one B→B field, the official reward, was missing from older trajectories and was recomputed from the stored reward information before the verdict. Besides the 924 sampled development states, 328/328 sampled Bank U states passed the QQ no-op, and the Bank U gate passed a further 135/135 no-ops (a different set from the holdout's 135 QQ no-ops, all of which also passed; Appendix N). Serving in separate processes closes

the prefix-cache channel (Patodiya, 2026). The same-precision run's acceptance counts are in Appendix D.6.

Determinism of the serving stack. Closed-loop determinism pairs, the same configuration on two nodes at different times, reproduced whole episodes token for token for dense BF16 paths at tensor parallelism 1 and 2 with 1 to 16 concurrent requests (e.g. 257/257 and 1520/1520 episodes). Nine arms that two studies ran independently at the same states were byte-identical. We did not test exactness under KV-cache preemption, or on other models, GPUs or vLLM versions. A deployed AWQ kernel path with fp16 dequantization was not replay-exact, and no reported outcome uses it.

User simulator. The served user simulator is Qwen3-30B-A3B-Instruct in every episode; a user-model label stored in the fork state files is outdated and should be ignored.

B End-to-End Success Comparison

The replacement gains G_{all}, G, H and F are not end-to-end comparisons. Table 1 gives the end-to-end comparison, computed separately on the same checkpoints, the same tasks and the same 20 nonce blocks as the forks.

Cell	Q / BF16 (%)	Q−BF16 [95% CI]	NI 3 pp	TOST ±3 pp
q4 GPTQ-W4	40.0 / 35.8	+4.2 [−2.4, +10.5]	pass	fail
q4 AWQ-W4	36.8 / 35.8	+0.9 [−5.8, +7.5]	fail	fail
q8 GPTQ-W4	30.1 / 30.4	−0.2 [−7.7, +6.8]	fail	fail
q8 AWQ-W4	31.5 / 30.4	+1.2 [−5.3, +7.6]	fail	fail
pool, all four		+1.5 [−2.7, +5.6]	pass	fail

Table 1: End-to-end closed-loop success, quantized minus BF16 (pp). Units are paired on nonce, and the crossed task×block bootstrap is drawn jointly across cells. Non-inferiority (NI) at a 3 pp margin passes for q4 GPTQ-W4 and the all-four pool. Equivalence (TOST at ±3 pp) passes nowhere. These comparisons therefore support neither a difference nor equivalence.

C Full Tables: Rescue, Damage and the Four Branches

Table 2 gives every cell of Figure 1, including the stress cells, with new-task intervals. Table 3 gives every cell of Figure 5, with $F-G$, $F-H$ and the $(\Delta_{\text{Q}}, \Delta_{\text{B}})$ summaries.

Cell	units	div.	r	d	M	G_{all} [fixed pool]	[new tasks]
q4 GPTQ-W4	85×20	1102	6.8	8.7	15.5	−1.9 [−3.7, −0.1]	[−6.8, +2.7]
q4 AWQ-W4	85×20	1439	9.8	10.1	19.9	−0.3 [−2.2, +1.6]	[−5.4, +5.2]
q8 GPTQ-W4	85×20	1473	10.4	7.5	17.9	+2.9 [+1.4, +4.4]	[−2.9, +9.2]
q8 AWQ-W4	85×20	1562	11.4	9.5	20.9	+1.9 [−0.6, +4.3]	[−3.9, +8.1]
W4 pool	85×20		9.6	9.0	18.6	+0.7 [−0.3, +1.6]	
q4 KV8	85×22	593	3.1	3.4	6.5	−0.3 [−1.7, +1.0]	[−3.0, +2.3]
q8 KV8	85×22	855	4.3	3.7	8.1	+0.6 [−0.4, +1.6]	[−1.9, +3.0]
q4 GPTQ-W3^s	85×3	232	9.0	10.2	19.2	−1.2 [−7.0, +4.7]	[−8.6, +5.9]
q4 KV4-rot^s	85×3	233	11.8	7.1	18.8	+4.7 [−4.1, +13.5]	[−3.1, +12.9]
q8 GPTQ-W3^s	85×3	242	10.2	8.2	18.4	+2.0 [−12.5, +16.4]	[−6.3, +11.8]
q8 KV4-rot^s	85×3	239	11.8	8.6	20.4	+3.1 [−14.5, +20.8]	[−5.9, +12.9]
q8 RTN-W3^s	85×3	255	3.1	3.5	6.7	−0.4 [−3.8, +3.0]	[−4.7, +3.9]

Table 2: Replacing Q's first tool-structure-divergent action while Q continues, τ^2-bench retail/airline panel (pp). Denominator: the full pool of tasks × complete blocks; units without a Bank P state count as zero. div. = Bank P states. $G_{\text{all}} = r - d$ is a local replacement effect, *not* the end-to-end success difference (Appendix B). Fixed pool: block bootstrap (≥10 blocks) or t over block means; new tasks: crossed task×block bootstrap. ^{s}stress cell, 3 blocks.

D Supplementary Results for the Main Findings

This appendix gives in full the analyses that the main text summarises.

D.1 Upgrade options within cells

On average over W4 cells, no option has a detectable gain. Pooled over the four W4 cells (Figure 5), $G = +0.5$, $H = +0.2$ and $F = +0.7$ pp, and every interval covers zero, as do their differences.

Within cells, the paired gains differ. In q8 GPTQ-W4, $G = +3.3$ [+1.6, +5.0] pp exceeds $H = +0.4$ ($G - H = +2.9$ [+1.3, +4.7]), and $F = +5.4$ exceeds G ($F - G = +2.1$ [+0.2, +4.0]). In q8 AWQ-W4, $G - H = +3.4$ [+1.5, +5.2] and $F - G = -1.5$ [−4.2, +0.8]. In q4 GPTQ-W4, $G = -2.9$ and $G - H = -3.6$ [−6.6, −0.6]. In the stress cell q8 RTN-W3, whose own success is low (three blocks, t interval), $H = +15.3$ and $F = +18.4$ against $G = -0.4$ ($G - H = -15.7$ [−27.5, −3.9]). In q4 KV8, B continuing after Q's action lowers success ($H = -2.5$ [−4.5, −0.5], unadjusted). These within-cell orderings are exploratory: they hold on the fixed 85 tasks but are not confirmed on new tasks (Table 5) or under a second simulator (Appendix D.4), and cells select different states.

The $G - H$ family and Figure 2. In Figure 2, G, H, F and the pool show fixed-pool 95% (thick; block bootstrap) and new-task 95% (thin; crossed) intervals, none multiplicity-adjusted. For

Cell	states	G (this step)	H (next step on)	F (from here on)	$G - H$	$F - G$	$F - H$ (= Δ_B)	off-diag. %	rev. %	mean I
q4 GPTQ-W4	1102	−2.9 [−5.7, −0.2]	+0.7 [−1.6, +3.0]	−2.5 [−5.3, +0.2]	−3.6 [−6.6, −0.6]	+0.5 [−1.2, +2.2]	−3.2 [−6.4, +0.2]	25.0	2.6	+0.3
q4 AWQ-W4	1439	−0.3 [−2.6, +1.9]	+0.9 [−1.1, +2.9]	−0.8 [−3.3, +1.6]	−1.3 [−3.0, +0.6]	−0.4 [−2.4, +1.5]	−1.7 [−3.6, +0.4]	29.3	3.1	+1.3
q8 GPTQ-W4	1473	+3.3 [+1.6, +5.0]	+0.4 [−1.2, +1.9]	+5.4 [+3.5, +7.2]	+2.9 [+1.3, +4.7]	+2.1 [+0.2, +4.0]	+5.0 [+3.2, +6.8]	22.9	2.7	−1.7
q8 AWQ-W4	1562	+2.1 [−0.6, +4.7]	−1.3 [−3.5, +0.9]	+0.6 [−1.5, +2.7]	+3.4 [+1.5, +5.2]	−1.5 [−4.2, +0.8]	+1.9 [−0.1, +3.8]	30.5	4.2	+0.3
W4 pool	5576	+0.5 [−0.7, +1.7]	+0.2 [−1.3, +1.6]	+0.7 [−0.6, +2.1]	+0.4 [−0.8, +1.7]	+0.2 [−1.0, +1.2]	+0.5 [−1.0, +2.1]	26.9	3.2	+0.0
q4 KV8	593	−1.0 [−5.3, +3.2]	−2.5 [−4.5, −0.5]	−1.2 [−4.7, +2.3]	+1.5 [−3.1, +6.0]	−0.2 [−3.5, +3.7]	+1.3 [−2.5, +4.9]	15.7	1.2	−2.4
q8 KV8	855	+1.3 [−0.9, +3.4]	−0.4 [−2.1, +1.5]	+2.5 [+0.0, +5.1]	+1.6 [−0.6, +3.9]	+1.2 [−0.5, +3.2]	+2.8 [+0.3, +5.5]	14.9	1.2	−1.5
KV8 pool	1448	+0.1 [−2.8, +3.0]	−1.4 [−2.7, −0.1]	+0.6 [−1.8, +3.1]	+1.6 [−1.2, +4.1]	+0.5 [−1.6, +2.8]	+2.1 [−0.2, +4.2]	15.3	1.2	−1.9
q4 GPTQ-W3^s	232	−1.3 [−7.7, +5.0]	+4.3 [−10.7, +19.4]	+1.3 [−1.9, +4.4]	−5.6 [−15.0, +3.7]	+2.6 [−0.6, +5.8]	−3.0 [−15.1, +9.1]	31.5	3.9	+1.7
q4 KV4-rot^s	233	+5.2 [−4.5, +14.9]	+9.4 [+5.1, +13.7]	+12.4 [+11.0, +13.9]	−4.3 [−18.3, +9.8]	+7.3 [−2.0, +16.5]	+3.0 [−1.9, +8.0]	33.0	1.7	+2.1
q8 GPTQ-W3^s	242	+2.1 [−13.3, +17.2]	+0.4 [−7.4, +8.2]	+2.1 [−7.4, +11.4]	+1.7 [−9.1, +12.3]	+0.0 [−11.0, +11.1]	+1.7 [−0.1, +3.4]	23.1	4.5	+0.4
q8 KV4-rot^s	239	+3.3 [−15.4, +21.9]	−0.8 [−18.2, +16.3]	+4.6 [−16.0, +25.0]	+4.2 [+2.5, +5.8]	+1.3 [−1.8, +4.3]	+5.4 [+0.8, +10.0]	31.0	4.6	−2.1
q8 RTN-W3^s	255	−0.4 [−3.8, +3.0]	+15.3 [+0.7, +29.9]	+18.4 [+16.7, +20.1]	−15.7 [−27.5, −3.9]	+18.8 [+13.8, +23.9]	+3.1 [−13.0, +19.2]	29.0	0.4	−3.5

Table 3: Three replacement gains and their paired differences on each cell's Bank P states (pp; same denominator in every column). $G = Y_{BQ} - Y_{QQ}$, $H = Y_{QB} - Y_{QQ}$, $F = Y_{BB} - Y_{QQ}$. Off-diag.: share of states with $\Delta_Q \neq \Delta_B$; rev.: strict reversals. Fixed-pool 95% intervals (block bootstrap over 20–22 complete blocks; t over 3 blocks for ^{s}stress cells, which cannot be pooled). New-task intervals are in Table 5. Within a row, contrasts are paired and identified; between rows, descriptive.

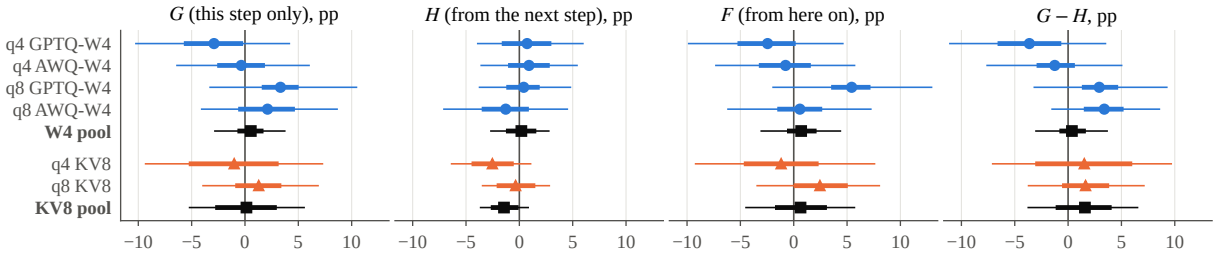


Figure 5: Replacement gains G, H, F and $G - H$ per core cell (pp; each cell's Bank P divergent states). Thick: fixed-pool 95% (block bootstrap, 20–22 blocks); thin: new-task 95% (crossed task×block). Squares: equal-weight pools. Within a row, contrasts are paired; between rows, descriptive. Stress cells, $F - G$, $F - H$ and (Δ_Q, Δ_B) summaries: Table 3.

the four cells' $G - H$, which form the Holm family ($m = 4$), thick is the plain 95% t interval and thin the Bonferroni simultaneous 98.75% interval; filled markers are significant after Holm adjustment, open markers are not. The stress cell has a t interval over three blocks, and the BFCL configuration an item-bootstrap 95% interval. The plain 95% t intervals are [+1.0, +4.8] (q8 GPTQ-W4, $G - H = +2.9$, Holm-adjusted $p = .013$) and [+1.3, +5.5] (q8 AWQ-W4, +3.4, $p = .012$). In q4 GPTQ-W4 (−3.6) the plain interval [−6.9, −0.3] excludes zero, but the Holm-adjusted $p = .069$; q4 AWQ-W4 (−1.3) has $p = .199$. Two further cautions apply. A significant result in one model and a non-significant one in the other is not a significant difference between models, and the cells select different divergent states. $G - H > 0$ shows that replacing the action beats handing over the continuation, not that replacing the action beats QQ.

Supplementary configuration. For Qwen3-14B GPTQ-W4 on BFCL (one configuration, one run per item, 471 divergent items; exploratory), action and continuation are complementary: keeping Q's action, a BF16 continuation adds $H = +0.4$ [−1.1, +1.9] pp; after B's action has replaced Q's, the BF16 continuation adds a further $F - G = +3.4$ [+1.3, +5.6] pp.

A BF16 reference action can lower success. In q4 GPTQ-W4, B's action lowers success under Q's continuation on the fixed task pool ($G = -2.9$ [−5.7, −0.2] pp, unadjusted; the new-task interval covers zero), with a similar point estimate under B's ($\Delta_B = -3.2$ [−6.4, +0.2]).

D.2 Off-diagonal share per cell and environment

Figure 6 gives the off-diagonal share per cell; the pooled (Δ_Q, Δ_B) tables are in Figure 9. Because replay is exact, each off-diagonal state is a reproducible $I \neq 0$ on the success scale at that state under this simulator, not replay noise; it does not by itself indicate a distinct interaction mechanism (Appendix F). Cohen's κ between Δ_Q and Δ_B, against a marginal-preserving re-pairing of states, is 0.29 in the W4 pool; it measures how well the two effects agree across states, not whether the interaction is significant (Appendix G.1). Per W4 cell the share is 22.9–30.5%. Strict sign reversals are 1.2–4.2% of states in each W4 and KV8 cell, or

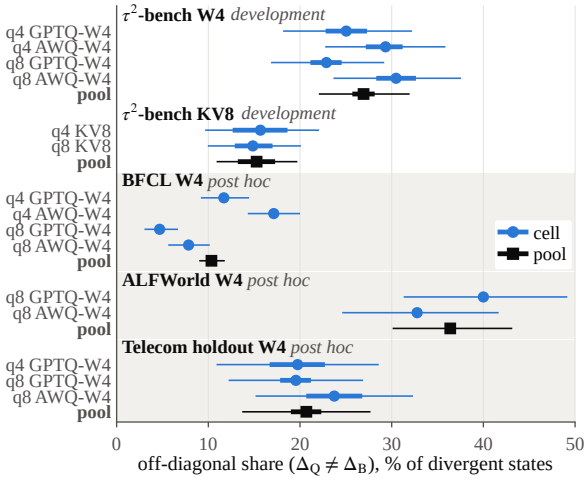


Figure 6: Same action, different value: per-cell share of divergent states with $\Delta_Q \neq \Delta_B$. Thick: fixed-pool 95% (development: block bootstrap; holdout: block t); thin: new-task 95% (crossed; BFCL/ALFWorld: item bootstrap). Pools: equal weight (holdout: pre-registered weights). Shaded groups are post hoc. Cross-cell comparisons are descriptive.

8–14% of the off-diagonal mass. In BFCL multi-turn and ALFWorld (post hoc; one run per item; Table 8), the off-diagonal share is 4.7–17.1% in the four BFCL W4 cells (pool 10.3%) and 32.8–40.0% in the two q8 ALFWorld W4 cells (pool 36.4%). Off-diagonal states therefore do not require an LLM user simulator, and their share differs by environment. The net replacement gain on these benchmarks covers zero except for BFCL q4 AWQ-W4, where it is -3.8 $[-6.5, -1.3]$ pp. The value of a fixed replacement policy is $E[G]$, $E[H]$ or $E[F]$, not $E[I]$. All 16 outcome patterns are in Appendix E.

D.3 Telecom: development tasks and the holdout's four branches

The holdout protocol, its exclusions and strata are in Appendix N; all 12 blocks were complete, and 1200 of the 1224 units were analysed after 2 structural task×cell exclusions. The cells were q4 GPTQ-W4, q8 GPTQ-W4 and q8 AWQ-W4, weighted 1/2, 1/4, 1/4; the block-t 90% interval of the primary endpoint is $[-6.08, -3.05]$. On 80 telecom development tasks (6 blocks; fixed-pool 95%), the two q8 holdout cells give $G_{\text{all}} = +2.7$ $[-4.7, +10.1]$ (GPTQ-W4) and -0.6 $[-4.9, +3.7]$ pp (AWQ-W4), against -5.6 and -6.6 on the holdout; q4 has no telecom development forks. A post hoc four-branch analysis of the 1180 holdout divergent states (Table 19) gives $G = -4.6$ $[-7.8, -1.5]$, $H = -2.8$ $[-6.3, +0.6]$, $F = -2.5$ $[-6.2, +1.3]$, $G - H = -1.8$ $[-4.8, +1.1]$ and mean $I = -5.0$ $[-9.6, -0.4]$ pp (crossed 95%), unlike the near-zero development W4 pool, with 20.7% of states off the diagonal (Figure 6). This G conditions on divergence and nearly equals G_{all} because 1180 of 1200 units diverge. In an exploratory analysis of the development data, the top 20% of tasks hold 57–71% of all flips in the W4 and KV8 cells (Appendix K).

D.4 Ordinary states and the user simulator

Ordinary states turn over less, mostly through text under our simulator. At ordinary states, replacing Q's action changes the outcome less often than at first tool-structure divergences (a descriptive comparison of different state populations, with no interval on the difference). Over blocks f01–f03 (task-equal; t intervals over block means), M per sub-bank is 7.8–9.0% for the q4 W4 cells in U1 and U2B, 4.7–9.0% for the q8 W4 cells and at most 4.7% for KV8. The net gain lies within ± 2.8 pp (U1, U2B) and ± 5.0 pp (U2Q), and every interval covers zero (Appendix J). Identical actions never change the outcome, as determinism implies. Reused arms rest on canonical equality. All 4,114 such reuses were rendered with each model's chat template; 4,113 are token-identical. The remaining pair differs only in JSON key order, and its two cross arms, measured separately, gave the reused outcomes. After deduplication, 150 of the 208 outcome-changing cell×state rows (≈72.1%) are class (ii), the same tool structure with different text; 26 involve a different tool name, call count or argument (Table 11). The rows are not independent states: the three blocks repeat the same tasks, and each U2B/U2Q state appears once per cell of a model. This is a *composition share*: per instance, text, tool/argument and act/talk differences flip the outcome in 14.4%, 18.2% and 18.9% of rows. Text dominates because it is common (1045 of 1357 differing rows), not because each text difference matters more. Among simple content features, only an above-median length difference between the two texts co-occurs with outcome change (in all four U1/U2B × model groups); question, confirmation-request and new-entity flags show no consistent association. These are untested regular-expression features. Bank P cannot see these differences by design.

Sensitivity to the user simulator. Text actions act through the user simulator, and changing the simulator moves τ-bench outcomes by 9–15 points (Seshadri et al., 2026; Jain and Sandhu, 2026). We froze a control before running it: the 39 deduplicated class-(ii) flips of block f01 and 39 matched non-flip states. Their BQ and QQ branches were each re-run three times with a different user simulator, Sonnet 5.5, answering through a file relay from Claude Code subagents (reproducibility limits in the Limitations). Judged by the sign of the replicate mean, only 7 of the 39 flips keep their direction (95% CI 0.075–0.335). Six reverse, and 26 vanish (15 both succeed, 2 both fail, 9 ties). Among the matched non-flip states, 11 of 39 become flips (0.15–0.45). Replicates of the same arm disagree in 41% of flip arms and in 21–26% of control arms. **The Bank U text-difference outcome labels did not reproduce reliably under this alternative relay setting.** The control changes three things at once (simulator model, sampling, relay interface), so it cannot say which of them changed the labels.

Sensitivity of the four-branch results. Before running, we sampled 80 Bank P states without outcome information (20 per W4 cell, stratified by domain and block) and re-ran all four arms five times each with Sonnet 5.5 as the user, one isolated single-request call per user turn (7896 calls; all integrity checks passed). Per-state arm outcomes track the original single runs (correlation 0.63–0.68). Pooled over cells, $G = -3.0$ $[-10.2, +4.8]$, $H = +5.5$ $[+0.0, +11.2]$ and $G - H = -8.5$ $[-14.8, -2.8]$ pp under Sonnet, against $G - H = -2.5$ $[-11.2, +6.2]$ on the same states under the original simulator (change -6.0 $[-16.2, +4.2]$). With 20 states per cell, this control cannot resolve the few-pp contrasts of Table 3 (on this sample even the original simulator gives $G - H = 0.0$ in both q8 W4 cells), so it neither confirms nor reverses the within-cell orderings. Per-cell estimates under both simulators are in Figure 10, and an exploratory mean squared interaction under Sonnet is in Appendix G.2.

Bank P and Bank U measure different quantities. Bank P skips text-only differences by design, yet these make up 150 of the 208 outcome-changing rows at ordinary states, and their labels did not reproduce under a second simulator (Appendix J; §4). Including text actions requires a simulator control before their consequences are reported as a property of the action.

D.5 Token-level analysis

Following Lotfi et al. (2026), we scored every Bank U f01 state at the token level (1140 cell $\times$ state rows, 58 flips), teacher-forcing both models on the generated ids of a_{B} and a_{Q}. We record truncated top-20 entropy, the top-1 margin and a top-20-union KL, not full entropy or KL. All analyses are exploratory, and f01 is development data.

Uncertainty is associated with divergence. Q's entropy at the start of its action separates states where B and Q act differently from states where they act identically, with AUC 0.66–0.91 per sub-bank and model. Over all ordinary states, flips also sit at lower Q margins (U1: AUC 0.31 [0.22, 0.40] for q4, 0.24 [0.13, 0.36] for q8; stratified).

Among divergent states, it does not reliably identify consequences. Restricting to states where the actions differ removes the "any difference at all" signal. In U1 the signal is not detectable: no Q-visible statistic has an interval excluding 0.5 (21 and 9 flips; f01). Only small sub-banks keep a signal: q4 U2B (mean entropy over a_{Q}, AUC 0.73 [0.53, 0.89]) and q4 U2Q (entropy and margin at the action start, AUC 0.90 and 0.10; 10 states, 4 flips). At the first divergent token, pooled entropy and KL do not separate flips from non-flips except in q4 U2Q (KL AUC 0.80 [0.65, 0.93], entropy 0.25 [0.07, 0.48]; 4 flips). Per cell, two further unadjusted intervals exclude 0.5, in opposite directions (U1 q4 GPTQ-W4 entropy 0.30; U2B q4 KV8 KL 0.10). No class of first divergent token reproduces the marker enrichment of Lotfi et al. (2026). The largest q4 enrichment is ordinary word choice (rate ratio 1.89, Fisher $p = .019$, not corrected). On development data, uncertainty is associated with *where* quantization changes the action more strongly than with *which* changes matter. The frozen trigger of §6 uses held-out blocks f02/f03 (1020 U1 rows).

D.6 Same-precision reference per cell

Per model (final analysis set, $K = 6$), the turnover difference is $\Delta M = +0.6$ $[-3.1, +4.3]$ pp in q4 and -2.5 $[-5.6, +0.5]$ in q8 (fixed pool, 95%), with crossed intervals $[-5.5, +6.9]$ and $[-7.3, +2.7]$. Within ± 3 pp, equivalence is shown for no per-model ΔM or ΔG_{all}; the q8 off-diagonal difference ($+0.40$, 90% $[-2.12, +2.92]$) is the only

per-model contrast inside the margin. The protocol originally mapped "interval includes 0 and $|\Delta M| \leq 3$ pp" to "similar"; after the interim result had been seen, an amendment replaced that reading by two one-sided tests, which are stricter, and changed no estimate. Acceptance: every state in both designs passed Q replay and every branch passed the prefix and environment checks; QQ no-op samples passed 265/265 (precision) and 242/242 (same precision), and no directory was quarantined. At the unit level, κ is 0.43 in q4 and 0.61 in q8; at the task level, cross-design agreement at matched noise is 0.765 of within-design replication.

W3 stress check (§5; $K = 3$): every state passed Q replay, QQ no-op samples passed 31/31 and 39/39, and no directory was quarantined. All branches passed the prefix and environment checks except one of 774 in q8 same precision: a Q′Q′ branch that ended in an agent-side context overflow, for which the harness records no prefix comparison; its environment hash and first replayed prompt match the frozen state, and by the protocol it counts as a failure, not an exclusion.

Equivalence tests (W4). The difference in turnover is $\Delta M = M - M' = -0.98$ pp (90% CI $[-3.26, +1.30]$). No precision-specific excess was detected, but equivalence within ± 3 pp is not shown either (two one-sided tests, $p = 0.067$), and the same holds for the net gain ($\Delta G_{\text{all}} = -1.57$ $[-5.17, +2.04]$, $p = 0.230$) and, descriptively, for the off-diagonal share (-1.210 $[-4.09, +1.67]$, $p = 0.133$, as the mean of per-block ratios, which the interval and test use; -1.216 as the ratio of block sums).

Overlap details (W4). Assuming that the two flip sets of Figure 3b are independent within cell and block would give 30 units flipping under both, against 107 observed and 76 [69, 84] under the within-task block shuffle. When B also takes over the continuation, rescue sets under the continuation-only contrast agree with $\kappa = 0.20$, and 70% of continuation-dependent units are continuation-dependent under one substitution only.

Full substitution (W4). Under full substitution B flips more units than Q′ (193 vs. 149; $+4.3$ $[+0.4, +8.3]$ pp, one of several contrasts); the net difference comes only from substitution-specific units and is not resolved ($\Delta G = -1.6$ $[-6.2, +3.0]$ pp). Only value measures on the same units reveal this structure.

W3 stress pair. The protocol's complete W3 panel (Q GPTQ-W3, Q′ a second W3 calibration draw; q4 and q8, $K = 3$ blocks q3–q5, 510 units) is reported next to W4, never pooled with it. The same-precision substitution produces 17.3 pp of turnover, the BF16 substitution 18.8 pp; off-diagonal shares are 27.6% and 27.3%. $\Delta M = +1.6$ pp (95% $[-0.66, +3.80]$) is unresolved: the preregistered ± 3 pp equivalence test fails ($p = 0.055$); that the 90% interval lies just above zero does not change this. ΔG_{all} is equivalent within ± 3 pp only in the preregistered pooled population ($+1.2$, $p = 0.017$); per model it is $+3.1$ (q4) and -0.8 pp (q8). The off-diagonal difference is -0.288 (ratio of block sums) or -0.307 (mean of per-block ratios, used by the interval and test: 90% $[-11.0, +10.4]$, $p = 0.27$). On the action side (G), rescued and harmed task×block units overlap ($\kappa = 0.60, 0.68$); the designs may first diverge at different decisions, so this is unit overlap, not a same-state replication. Of 122 units whose G outcome flips, 62 flip under both substitutions, 34 only under BF16 and 26 only under the other W3 checkpoint: 49% respond to only one. Flip volume, affected units and net upgrade value must be measured separately.

Net values on the same-precision blocks. The reference changes how B's consequences are read, not what they are: on 20 blocks B's action is worth $G_{\text{all}} = +2.9$ pp in q8 and -1.9 pp in q4 GPTQ-W4 (§4). On the same-precision blocks, B's and Q′'s net gains (-0.7 $[-3.9, +2.5]$ and $+0.9$ $[-3.1, +4.8]$ pp) are no conclusion about value; the next comparison asks it directly.

Common-state comparison: details. Above, each design selects its own divergent states. A pre-registered comparison removes that difference: on 510 ordinary q8 states (85 tasks × 6 blocks, sampled by a fixed rule), the current action comes from B, from Q (GPTQ-W4, first checkpoint) or from Q′ (second checkpoint), and B or Q continues. $K(c) = V(\text{B}, c) - V(\text{Q}', c)$ asks what B's action adds beyond another W4 action. A frozen rule claims value only if both intervals exclude zero (Holm), equivalence only if both lie within ± 3 pp (two one-sided tests). The design first ran three blocks, and that first look was unresolved (under Q's continuation $K = -0.4$ pp,

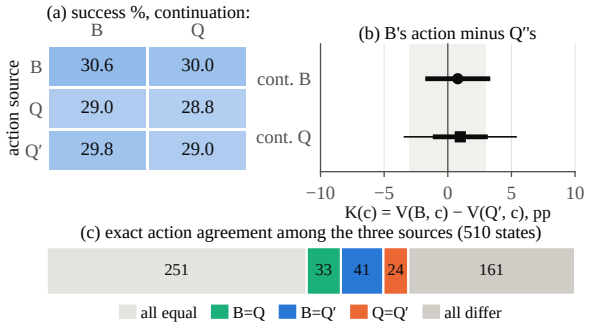


Figure 7: Common-state action sources (q8; 510 preregistered ordinary states, 6 blocks). (a) Success (%) by action source and continuation. (b) $K(c)$; thick: task-cluster bootstrap 95%, thin: t (block means) 95%; shaded: ±3 pp; two-look sensitivity intervals: see text. (c) Exact agreement of the three actions.

95%: t [−9.3, +8.5], bootstrap [−3.5, +2.7]; under B's +2.4 pp, [−3.5, +8.2], [−0.8, +5.5]). Three more blocks were then added. That decision followed the unresolved first look, so a dated amendment fixed the extension before any new outcome existed, and a two-look sensitivity analysis (Pocock levels from a normal approximation: about 97% intervals for a difference and 94% for equivalence) is reported next to the ordinary intervals. The Pocock constant is not calibrated for small-k t intervals, the task bootstrap or the joint rule, so it is a sensitivity analysis, not an exact error guarantee. After the fixed extension (six blocks), B's action is not shown to be worth more than Q′'s, and equivalence is not established. Under Q's continuation, $K = +1.0$ pp (95%: t [−3.5, +5.4], bootstrap [−1.2, +3.1]; two-look 97%: [−4.2, +6.2], [−1.4, +3.5]). Under B's, $K = +0.8$ pp (95%: [−1.8, +3.3] for both; two-look 97%: [−2.2, +3.8], [−2.0, +3.7]). For equivalence under B's continuation, the ordinary 90% intervals lie within ±3 pp (t [−1.2, +2.8], bootstrap [−1.4, +2.9]) and the two-look 94% intervals do not (t [−1.6, +3.2]); because the extension followed the first look, we do not claim equivalence. Without the 26 states at which earlier studies had already produced outcomes, K is +0.5 and +0.8 pp, also not resolved. Under Q's continuation, B's action flips $M = 7.1\%$ of outcomes (rescue 4.1, damage 2.9) for a net $\Delta = +1.2$ pp (95%: t [−1.3, +3.6]); Q′'s flips 8.0% (4.1, 3.9) for +0.2 pp ([−2.3, +2.7]; within ±3 pp by two one-sided tests). A second W4 checkpoint's action departs from Q's at least as often as B's does (agreement with Q's action 53.9% against 55.7%; B with Q′ 57.3%; Figure 7c). The interaction of K with the continuation is −0.2 pp (95%: t [−4.5, +4.1], bootstrap [−2.7, +2.4]; secondary, not resolved). *q4, a separate prospective analysis.* The same design on Qwen3-4B planned three blocks. The frozen quarantine rule removed f01 (below), leaving 170 states. $K = -1.2$ pp under Q's continuation (bootstrap 95% [−5.9, +3.5]) and −2.4 pp under B's ([−6.5, +1.2]), both not resolved. With two blocks the t interval has one degree of freedom. Under Q's continuation the two block means happen to coincide, so it collapses to a point. That is a coincidence of two values, not a precise measurement or zero variance, and the reading rests on the bootstrap. *Identity checks.* Across both models, 1,238 of 1,240 reference and implicit identity checks passed. Both failures are one q4 state. There the substituted action was semantically identical to the original after parsing, but its token sequence differed in the order of keys inside a tool-call argument. The canonical comparison treats such actions as equal, but the chat template renders the order, so the next prompt differed and the continuation could not be a replay. Every check with a token-identical action passed (869/869). As the frozen rule requires, the block was removed. Changing outcomes is thus not improving them; this does not make precision irrelevant or the two actions equivalent.

E All 16 Outcome Patterns and New-Task Intervals

Table 4 lists every four-branch pattern per cell; `0000` (39–58%) and 1111 (6–20%) dominate. Table 5 gives the new-task intervals for Table 3.

F Outcome Classes and a Label-Permutation Reference

Everything in this section is exploratory and post hoc. It uses the four outcomes of each divergent state in the order BB, BQ, QB, QQ; k is the number of successful arms.

Five classes. The 16 patterns fall into five classes: unchanged (`0000`, `1111`); action only (`1100`, `0011`: the outcome follows the action under both continuations); continuation only (`1010`, `0101`); one arm differs from the other three (k odd); strict reversal (`1001`, `0110`). Changing the action changes the outcome under at least one continuation exactly in the action-only, one-arm and

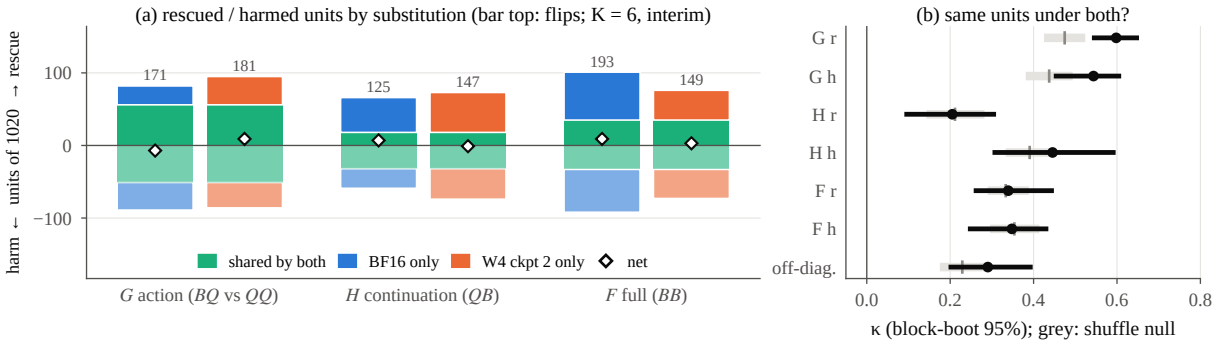


Figure 8: Which units the two substitutions rescue and harm (q4 and q8 GPTQ-W4 pooled; 1020 units; final analysis set, $K = 6$ blocks; descriptive). (a) Per contrast (G: action side, BQ vs. QQ; H: continuation side, QB vs. QQ; F: full substitution, BB vs. QQ) and substitution (B, or the second W4 checkpoint in the B role): rescues above zero and harms below, split into units shared by both substitutions and substitution-specific units; the bar-top number is the flip count and the diamond the net. (b) Cohen's κ of each set between the two substitutions (block-bootstrap 95%) against a shuffle null (grey: within task and QQ outcome for rescue and harm sets, within task for the off-diagonal set). Rescue under one substitution and harm under the other cannot co-occur, because the QQ episode is shared.

$(Y_{BB}, Y_{BQ}, Y_{QB}, Y_{QQ})$	q4 AWQ	q4 GPTQ-W3	q4 GPTQ	q4 KV4r	q4 KV8	q8 AWQ	q8 GPTQ-W3	q8 GPTQ	q8 KV4r	q8 KV8	q8 RTN-W3
0000	634 (44.1%)	105 (45.3%)	472 (42.8%)	92 (39.5%)	286 (48.2%)	767 (49.1%)	130 (53.7%)	766 (52.0%)	113 (47.3%)	497 (58.1%)	141 (55.3%)
0001	55 (3.8%)	7 (3.0%)	29 (2.6%)	3 (1.3%)	14 (2.4%)	66 (4.2%)	2 (0.8%)	36 (2.4%)	8 (3.3%)	15 (1.8%)	3 (1.2%)
0010	57 (4.0%)	13 (5.6%)	54 (4.9%)	13 (5.6%)	11 (1.9%)	38 (2.4%)	8 (3.3%)	34 (2.3%)	10 (4.2%)	15 (1.8%)	20 (7.8%)
0011	58 (4.0%)	6 (2.6%)	71 (6.4%)	4 (1.7%)	37 (6.2%)	30 (1.9%)	9 (3.7%)	49 (3.3%)	3 (1.3%)	39 (4.6%)	0 (0.0%)
0100	50 (3.5%)	6 (2.6%)	36 (3.3%)	9 (3.9%)	8 (1.3%)	53 (3.4%)	8 (3.3%)	45 (3.1%)	13 (5.4%)	20 (2.3%)	1 (0.4%)
0101	21 (1.5%)	4 (1.7%)	17 (1.5%)	4 (1.7%)	11 (1.9%)	36 (2.3%)	7 (2.9%)	12 (0.8%)	5 (2.1%)	5 (0.6%)	2 (0.8%)
0110	22 (1.5%)	5 (2.2%)	14 (1.3%)	2 (0.9%)	2 (0.3%)	33 (2.1%)	4 (1.7%)	16 (1.1%)	2 (0.8%)	5 (0.6%)	1 (0.4%)
0111	41 (2.8%)	5 (2.2%)	21 (1.9%)	7 (3.0%)	13 (2.2%)	51 (3.3%)	6 (2.5%)	31 (2.1%)	7 (2.9%)	7 (0.8%)	6 (2.4%)
1000	51 (3.5%)	8 (3.4%)	32 (2.9%)	19 (8.2%)	16 (2.7%)	67 (4.3%)	10 (4.1%)	75 (5.1%)	16 (6.7%)	28 (3.3%)	32 (12.5%)
1001	23 (1.6%)	4 (1.7%)	15 (1.4%)	2 (0.9%)	5 (0.8%)	33 (2.1%)	7 (2.9%)	24 (1.6%)	9 (3.8%)	5 (0.6%)	0 (0.0%)
1010	18 (1.3%)	5 (2.2%)	13 (1.2%)	9 (3.9%)	4 (0.7%)	17 (1.1%)	5 (2.1%)	17 (1.2%)	3 (1.3%)	3 (0.4%)	20 (7.8%)
1011	36 (2.5%)	9 (3.9%)	33 (3.0%)	9 (3.9%)	8 (1.3%)	32 (2.0%)	3 (1.2%)	19 (1.3%)	2 (0.8%)	11 (1.3%)	6 (2.4%)
1100	44 (3.1%)	3 (1.3%)	51 (4.6%)	9 (3.9%)	41 (6.9%)	43 (2.8%)	10 (4.1%)	82 (5.6%)	9 (3.8%)	46 (5.4%)	2 (0.8%)
1101	36 (2.5%)	7 (3.0%)	27 (2.5%)	3 (1.3%)	9 (1.5%)	38 (2.4%)	4 (1.7%)	23 (1.6%)	1 (0.4%)	11 (1.3%)	1 (0.4%)
1110	51 (3.5%)	9 (3.9%)	15 (1.4%)	10 (4.3%)	7 (1.2%)	65 (4.2%)	4 (1.7%)	34 (2.3%)	6 (2.5%)	10 (1.2%)	4 (1.6%)
1111	242 (16.8%)	36 (15.5%)	202 (18.3%)	38 (16.3%)	121 (20.4%)	193 (12.4%)	25 (10.3%)	210 (14.3%)	32 (13.4%)	138 (16.1%)	16 (6.3%)
n	1439	232	1102	233	593	1562	242	1473	239	855	255

Table 4: All 16 four-branch outcome patterns per cell: count (share of the cell's Bank P states). Pattern digits are $(Y_{BB}, Y_{BQ}, Y_{QB}, Y_{QQ})$. KV4r: Q/K-rotated KV4; W3 and KV4r cells are stress cells with 3 blocks.

reversal classes, and changing the continuation exactly in the continuation-only, one-arm and reversal classes. Both change it exactly on the off-diagonal states, so the two shares overlap and must not be added, and their gap equals action only minus continuation only: it comes from $k = 2$ states alone and restates the $k = 2$ arrangement below on all divergent states. Two named patterns follow the upgrade modes: joint rescue (1000: only the joint upgrade succeeds) and rescue undone ($Y_{BQ} = 1$, $Y_{QQ} = 0$, $Y_{BB} = 0$: B's action rescues under Q's continuation and B's takeover of the continuation undoes it). Table 6 gives the shares. The action-side gap holds in the τ^2-bench retail/airline W4 and KV8 pools, in the W4 same-precision fork and in the BFCL W4 pool. It is near zero on the telecom development tasks (+0.1 [−2.1, +2.4] pp, 6-block t) and in the q4 GPTQ-W3 and KV4-rot stress cells, and it is not resolved in the ALFWorld W4 pool (+3.4 [−1.8, +8.4]) or the W3 same-precision pair (+1.9 [−2.9, +6.7]). Its point estimate is negative in the q8 RTN-W3 stress cell (−7.8; 3-block t [−18.1, +2.4], new-task [−15.3, −1.2]) and, not shown, in ALFWorld q8 GPTQ-W3 (−19.2 [−27.3, −10.8]). In the ALFWorld and BFCL KV4 cells (not shown) Q almost never succeeds, so continuation-only states dominate there mechanically.

W4 pool details. In the τ^2-bench W4 pool, the gap between the action and continuation shares is +5.2 pp (new-task [+3.1, +7.7]): action effects are somewhat more common, and the continuation changes the outcome mostly where the action matters too. Of the 26.9% off-diagonal states, 23.8% have one arm differing from the other three and 3.2% are strict reversals; states where the swap helps more under Q ($I > 0$, 13.6%) and under B ($I < 0$, 13.4%) are about equally frequent, so

Cell	G	H	F	$G-H$	$F-G$	$F-H$
q4 GPTQ-W4	$[-10.3, +4.2]$	$[-4.0, +6.0]$	$[-9.9, +4.7]$	$[-11.2, +3.6]$	$[-4.7, +6.1]$	$[-11.1, +4.4]$
q4 AWQ-W4	$[-6.5, +6.1]$	$[-3.7, +5.5]$	$[-7.4, +5.8]$	$[-7.7, +5.1]$	$[-6.2, +5.4]$	$[-8.7, +4.7]$
q8 GPTQ-W4	$[-3.3, +10.5]$	$[-3.8, +4.9]$	$[-2.0, +13.0]$	$[-3.2, +9.3]$	$[-3.0, +7.1]$	$[-2.0, +12.5]$
q8 AWQ-W4	$[-4.1, +8.7]$	$[-7.1, +4.6]$	$[-6.2, +7.3]$	$[-1.6, +8.6]$	$[-7.6, +4.2]$	$[-3.8, +7.5]$
W4 pool	$[-2.9, +3.8]$	$[-2.7, +2.8]$	$[-3.1, +4.4]$	$[-3.1, +3.7]$	$[-2.8, +3.1]$	$[-3.5, +4.5]$
q4 KV8	$[-9.4, +7.3]$	$[-6.4, +1.1]$	$[-9.3, +7.7]$	$[-7.1, +9.7]$	$[-5.7, +5.3]$	$[-6.7, +9.5]$
q8 KV8	$[-4.0, +6.9]$	$[-3.5, +2.9]$	$[-3.5, +8.1]$	$[-3.8, +7.2]$	$[-2.5, +5.0]$	$[-3.1, +8.7]$
KV8 pool	$[-5.3, +5.6]$	$[-3.7, +0.9]$	$[-4.5, +5.8]$	$[-3.8, +6.6]$	$[-2.8, +3.9]$	$[-3.0, +7.3]$
q4 GPTQ-W3^s	$[-9.7, +6.6]$	$[-6.6, +14.5]$	$[-6.8, +9.4]$	$[-15.0, +4.0]$	$[-5.2, +10.5]$	$[-12.2, +6.6]$
q4 KV4-rot^s	$[-3.4, +14.2]$	$[+2.1, +17.0]$	$[+3.0, +21.6]$	$[-13.5, +7.0]$	$[-1.3, +17.1]$	$[-6.2, +12.2]$
q8 GPTQ-W3^s	$[-6.7, +12.1]$	$[-7.2, +8.1]$	$[-6.6, +11.8]$	$[-7.7, +11.2]$	$[-9.1, +9.1]$	$[-6.8, +11.0]$
q8 KV4-rot^s	$[-6.6, +13.8]$	$[-11.2, +8.5]$	$[-6.6, +15.9]$	$[-3.3, +12.2]$	$[-7.2, +9.9]$	$[-2.5, +14.6]$
q8 RTN-W3^s	$[-4.7, +3.9]$	$[+5.9, +25.9]$	$[+9.4, +28.2]$	$[-24.7, -7.1]$	$[+10.2, +27.8]$	$[-7.1, +13.3]$

Table 5: New-task generalisation: crossed task×block bootstrap 95% intervals for the estimates of Table 3 (pp; same Bank P states). These answer whether the effect would hold for new tasks drawn like the panel; the fixed-pool intervals of Table 3 treat the 85 tasks as fixed. ^{s}stress cell, 3 blocks.

the mean interaction is $+0.0$ $[-1.9, +2.0]$ pp.

A label-permutation reference. Given k, assigning the four outcomes to the four arms uniformly at random gives the off-diagonal share $E_0 = P(k \text{ odd}) + P(k = 2)/3$. In every cell the observed share minus E_0 equals $P(k = 2)\,[P(\text{reversal} \mid k = 2) - 1/3]$, so it can depart from E_0 only through strict reversals. In the W4 pool the share is 26.9% against $E_0 = 28.4\%$ (-1.4 $[-1.8, -1.0]$ pp, fixed pool, block bootstrap over 20 blocks; KV8: 15.3 against 18.9%), because reversals are 23.6% [19.2, 28.3] (task bootstrap) of $k = 2$ states instead of one third. This shows only that strict reversals are rarer than under random labelling. It does not show that the continuation controller is irrelevant, which would require $Y_{\mathrm{BB}} = Y_{\mathrm{BQ}}$ and $Y_{\mathrm{QB}} = Y_{\mathrm{QQ}}$, or that the interaction is zero: the reference itself allows $I \neq 0$ in most arrangements. Conversely, an off-diagonal state does not show a distinct interaction mechanism: if an arm succeeds when an action score plus a continuation score exceeds a threshold, exactly additive scores can make only BB succeed (`1000`), although they never produce a strict reversal.

The $k = 2$ arrangement. At $k = 2$ the two successes share the action in 56.4% of W4 states and 80.3% of KV8 states, against one third under random labelling. A statement that outcomes are mainly determined by the action holds only within this subset, which is 13.8% (W4) and 14.5% (KV8) of divergent states.

Tests that treat states as independent. Ignoring task and block clustering, the reversal share at $k = 2$ differs from one third (exact binomial $p = 8.5 \cdot 10^{-9}$ for W4, $5.7 \cdot 10^{-17}$ for KV8),

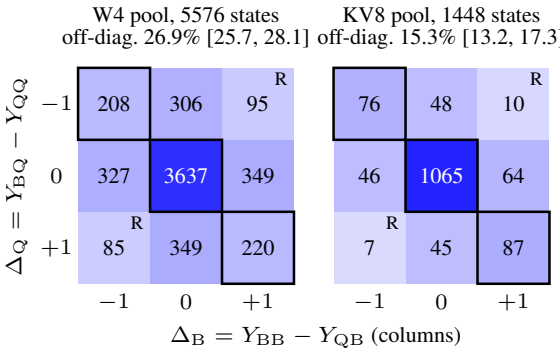


Figure 9: Does the action swap have the same effect under both controllers? Counts of (Δ_Q, Δ_B) over the Bank P states of the W4 cells (left) and the KV8 cells (right). Boxed: diagonal. R: strict reversals. Shading: share of states (power scale). Headers give the off-diagonal share as the equal-weight average over cells, the aggregation used throughout the paper except for the state-weighted menu oracle of Appendix L, with fixed-pool 95% intervals.

and the $k = 2$ arrangement differs from uniform (Monte Carlo multinomial $p < 10^{-4}$, the resolution of $R = 20000$ draws). States of one task and one block are correlated, so these p-values overstate the evidence; the main text relies on block and crossed intervals.

Task share of variance. An exploratory one-way ICC estimate with tasks as groups (moment estimator, not truncated) puts 0.17 [0.12, 0.22] of the state-level variance of I at the task level in the W4 pool (task bootstrap) and 0.00 in the KV8 pool. It does not separate a shared nonce (block) main effect, so the remainder cannot be called random noise.

Group	states	action changes	continuation changes	gap (pp)	unch.	act. only	cont. only	one arm	rev.	joint resc.	resc. undone
τ^2-bench W4 pool (4 cells)	5576	34.9 [33.6, 36.0]	29.6 [28.5, 30.7]	+5.2 [+4.4, +6.1]	62.4	7.9	2.7	23.8	3.2	4.0	4.8
τ^2-bench KV8 pool	1448	26.8 [25.0, 28.6]	17.0 [14.9, 19.1]	+9.8 [+8.0, +11.6]	71.5	11.6	1.7	14.1	1.2	3.0	2.3
Telecom q8 W4 pool	957	22.1 [18.2, 25.9]	21.9 [17.8, 26.1]	+0.1 [−2.1, +2.4]	74.8	3.2	3.1	18.1	0.7	3.2	4.2
Same-precision W4 (Q′ as B)	784	33.5 [29.8, 37.1]	28.7 [24.8, 32.2]	+4.9 [+2.9, +6.9]	64.0	7.4	2.5	22.5	3.6	2.3	5.8
BF16, same blocks	784	35.1 [32.6, 37.4]	27.8 [23.8, 31.6]	+7.3 [+4.9, +9.7]	62.0	10.2	2.9	22.4	2.6	5.0	3.8
q8 RTN-W3^s	255	29.8 [28.1, 31.5]	37.6 [26.0, 49.3]	−7.8 [−18.1, +2.4]	61.6	0.8	8.6	28.6	0.4	12.6	0.8
BFCL W4 pool^b	2296	18.4 [16.4, 20.5]	11.2 [9.8, 12.6]	+7.2 [+5.8, +8.7]	80.8	8.1	0.8	9.8	0.5	2.4	1.4
ALFWorld q8 W4 pool^b	234	44.9 [38.3, 52.4]	41.5 [34.3, 49.4]	+3.4 [−1.8, +8.4]	50.0	8.5	5.1	33.4	3.0	9.4	5.5

Table 6: Which intervention changes the outcome, in % of divergent states (pools: equal weight per cell). "Action changes": the outcome changes with the action under at least one continuation; "continuation changes": with the continuation controller under at least one action; gap = action only − continuation only. Fixed-pool 95% intervals: block bootstrap for the 20- and 22-block τ^2-bench W4 and KV8 pools; t over per-block ratios for the 6-block telecom and same-precision rows and the 3-block ^{s}stress cell; ^{b}one run per item, item bootstrap. Same-precision rows: Q′, a second GPTQ checkpoint of the same bit width, in the B role, and BF16 on the same blocks. New-task (crossed) intervals for the τ^2-bench W4 pool: §4 and above.

G Supplementary Four-Branch Analyses

G.1 Marginal-preserving pairing agreement control

For each cell, observed agreement is the diagonal share of the (Δ_Q, Δ_B) table, $p_o = \sum_k P(\Delta_Q = k, \Delta_B = k)$, so the off-diagonal share is $1 - p_o = P(I \neq 0)$. The control re-pairs states independently within a cell while keeping both marginals, $p_e = \sum_k p_Q(k)\, p_B(k)$, and Cohen's $\kappa = (p_o - p_e)/(1 - p_e)$. Pools use the paper's equal weight per cell over common complete blocks, with the same bootstrap draws as Table 3. This is a pairing agreement control, not a no-interaction null and not a chance baseline for controller dependence (§4). In the W4 pool the control expects 38.0% [36.7, 39.0] off-diagonal against the observed 26.9%, so $\kappa = 0.29$ [0.27, 0.32] (new-task [0.22, 0.36]); in the KV8 pool it expects 33.6% ($\kappa = 0.55$). The effects under the two controllers agree more than random cross-state pairing would. On BFCL and ALFWorld the observed shares also lie below the control (expected 16.2–32.4% and 42.9–46.4% in the W4 cells; κ 0.47–0.71 and 0.14–0.24; Table 8), and on the telecom holdout the pooled 20.7% compares with 25.7% expected ($\kappa = 0.19$; Table 19).

G.2 Mean squared interaction under the second simulator (exploratory)

On the 80 states of the Bank P simulator control (Appendix D.4), each arm has $n_a = 5$ replicates under Sonnet 5.5. Per state, $\hat{I}_s = \bar{Y}_{BQ} + \bar{Y}_{QB} - \bar{Y}_{BB} - \bar{Y}_{QQ}$ and $U_s = \hat{I}_s^2 - \sum_a \bar{Y}_a(1-\bar{Y}_a)/(n_a - 1)$, which is unbiased for I_s^2 if, given the state, the replicates of each arm are i.i.d. Bernoulli and the four arms are independent (no shared cache or seed). Pooled with equal weight over the four W4 cells, mean $\hat{I}_s^2$ is 0.110, the noise correction 0.060 and mean U_s 0.050, with 95% intervals [0.004, 0.107] over states and [0.001, 0.111] over task clusters. One cell is negative (q4 AWQ-W4, −0.025) and is kept, as are negative per-state values. The estimate is the mean squared probability-scale interaction over these 80 frozen states. It is not the share of states with interaction and not a variance, it is not comparable with the deterministic 26.9% of §4, and we make no significance claim.

Cell	states	off-diag. %	exp. %	κ [fixed pool]	κ [new tasks]
q4 GPTQ-W4	1102	25.0 [22.8, 27.3]	40.5	0.381 [0.325, 0.437]	[0.245, 0.520]
q4 AWQ-W4	1439	29.3 [27.2, 31.2]	38.5	0.238 [0.215, 0.261]	[0.138, 0.343]
q8 GPTQ-W4	1473	22.9 [21.1, 24.5]	36.3	0.369 [0.324, 0.409]	[0.235, 0.492]
q8 AWQ-W4	1562	30.5 [28.3, 32.6]	36.8	0.171 [0.128, 0.213]	[0.075, 0.274]
W4 pool	5576	26.9 [25.7, 28.1]	38.0	0.291 [0.265, 0.318]	[0.223, 0.359]
q4 KV8	593	15.7 [12.7, 18.6]	36.2	0.567 [0.487, 0.643]	[0.412, 0.705]
q8 KV8	855	14.9 [12.9, 17.0]	31.1	0.522 [0.462, 0.584]	[0.386, 0.656]
KV8 pool	1448	15.3 [13.2, 17.3]	33.6	0.546 [0.487, 0.604]	[0.436, 0.652]
q4 GPTQ-W3^s	232	31.5 [10.5, 52.6]	36.1	0.129 [−0.258, 0.534]	[−0.049, 0.347]
q4 KV4-rot^s	233	33.0 [17.6, 48.3]	38.0	0.131 [−0.016, 0.280]	[−0.034, 0.298]
q8 GPTQ-W3^s	242	23.1 [6.0, 40.2]	36.4	0.364 [0.190, 0.548]	[0.199, 0.532]
q8 KV4-rot^s	239	31.0 [19.0, 43.0]	37.7	0.179 [0.098, 0.251]	[0.032, 0.327]
q8 RTN-W3^s	255	29.0 [24.6, 33.5]	28.6	−0.016 [−0.233, 0.200]	[−0.101, 0.142]

Table 7: Pairing agreement control per cell (Bank P divergent states). Off-diag.: observed share with $\Delta_Q \neq \Delta_B$ (fixed-pool 95%); exp.: share expected under marginal-preserving within-cell re-pairing. Pools: equal weight per cell. ^{s}Stress cells (3 blocks; t intervals; flagged, never pooled). In q8 RTN-W3, $\Delta_Q = 0$ on 93.3% of states, so p_e is close to p_o and $\kappa \approx 0$.

G.3 Replacement and four-branch estimands on BFCL and ALFWorld (supplementary)

Table 8 gives the replacement quantities of §4 (r, d, M and G_{all} over all items, non-divergent items counting as zero) and applies the four-branch analysis to the external benchmarks, from the raw four-arm outcomes on divergent items (one run per item; item bootstrap within benchmark category).

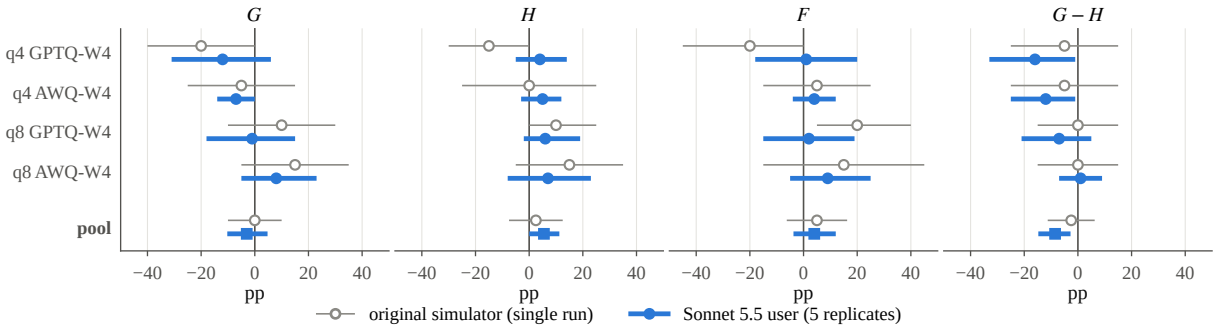


Figure 10: Bank P simulator control: G, H, F and $G - H$ on the 80 sampled Bank P states (20 per W4 cell), under the original greedy simulator (single runs; open circles) and with Sonnet 5.5 as the user (five replicates per arm; filled). pp; 95% intervals over states; pool: equal weight per cell. The sample is too small to test few-pp contrasts (Appendix D.4).

The four-branch columns are post hoc and not part of any pre-registered analysis. In the KV4 cells $Y_{\mathrm{QQ}} = 0$ on every state, so Δ_{Q} is almost constant, their off-diagonal mass is H and κ is near 0 by construction. In BFCL q4 GPTQ-W3, the base category was dropped whole by the QQ no-op rule, leaving 591 items and 540 states. Off-diagonal shares are 4.7–17.1% in BFCL W4 cells (pool 10.3%) and 32.8–40.0% in q8 ALFWorld W4 (pool 36.4%); pairing κ is in Appendix G.1. The ALFWorld GPTQ-W3 stress cell has $H = +38.5$ and $G - H = -34.6$ $[-43.5, -25.4]$ pp; KV4 cells reduce to H.

H Coverage Matrix

Table 9 lists the completed crossed-intervention coverage behind every Qwen3 Bank P and Bank U result; the coverage of the panels of §7 is in Appendix I. It counts tasks, states and complete blocks after the strict admission rule. Four of the five stress cells turn over 18.4–20.4% of units with three blocks; we keep them separate from W4, and W2 runs did not complete.

I Further Panels: Definitions, Provenance and Details

This appendix gives the details of the three panels of §7, which use ordinary BF16-sourced (U2B) states.

Contrasts. Methods are compared on shared states by $D_{cc} = Y_{\mathrm{QB}} - Y_{\mathrm{BB}}$ $(= -\Delta_{\mathrm{B}})$: Q's action against B's under the common BF16 continuation. Because BB is the shared BF16 original, two methods differ on the same states only through Y_{QB}. One pre-registered Holm family holds three contrasts on D_{cc}, with t intervals over block-mean differences: P1, QuaRot-W4 − GPTQ-W4; P2, SpinQuant-W4A4 − QuaRot-W4A4; P3, KIVI-2 − KV8. Intervals in §7 are fixed-pool 95% (t over block means, as fewer than 10 blocks are complete); new-task (crossed) intervals are in the analysis outputs.

Numerical quality. QuaRot-W4 against GPTQ-W4 is -0.4 $[-10.7, +9.9]$ pp (P1). The matched SpinQuant-W4A4 contrast is P2. Its third block rests on a one-time re-run of arms lost to a user-simulator overload; on the two blocks without it the contrast is $+2.4$ pp (strict view).

What local fidelity predicts (exploratory, post hoc). Added after the panel results were read. On each cell's admitted states we split the action swap under the common BF16 continuation into the action-difference rate $P(a_{\mathrm{Q}} \neq a_{\mathrm{B}})$ and the flip rate given a difference, whose product is the turnover M_{cc}; B's rescues $r_{cc} = P(Y_{\mathrm{BB}} = 1, Y_{\mathrm{QB}} = 0)$ and damages $d_{cc} = P(Y_{\mathrm{BB}} = 0, Y_{\mathrm{QB}} = 1)$ give $D_{cc} = d_{cc} - r_{cc}$. KL is the mean token-level KL of Q against B, teacher-forced on B's action. In QuaRot-W4A4 and SpinQuant-W4A4 the actions differ in 67.1 and 67.8% of states, a difference flips the outcome in 19.9 and 22.5%, and B's action rescues 8.2 and 8.6% and damages 5.1 and 6.7%. Over the eleven q8 cells, mean KL has Spearman $\rho = 0.99$ with the action-difference rate, 0.925 with M_{cc}, 0.81 with r_{cc}, 0.93 with d_{cc}, 0.46 with the flip rate given a difference ($p = .15$), -0.44 with D_{cc} and -0.42 with B's rescue share r_{cc}/M_{cc} (without the four unquantized path controls: 0.98, 0.64 and -0.52 for the difference rate, flip rate and D_{cc}); cells are not exchangeable draws, so these are descriptive. Within cells, KL's mean state-level AUC is 0.93 for a dif-

	all items					divergent items								
Cell	items	r	d	M	G_{all}	states	G	H	F	$G - H$	off-diag. %	exp. %	κ	rev. (n)
BFCL q4 GPTQ-W4	788	6.3	4.3	10.7	+2.0 [−0.3, +4.3]	573	+2.8 [−0.4, +5.9]	+1.4 [−0.7, +3.5]	+4.2 [+0.9, +7.2]	+1.4 [−1.8, +4.5]	11.7	27.4	0.574	3
BFCL q4 AWQ-W4	788	5.7	9.5	15.2	−3.8 [−6.5, −1.3]	642	−4.7 [−7.8, −1.5]	−1.1 [−3.6, +1.4]	−1.4 [−4.8, +1.7]	−3.6 [−6.7, −0.2]	17.1	32.4	0.471	8
BFCL q8 GPTQ-W4	788	2.9	2.3	5.2	+0.6 [−0.9, +2.3]	533	+0.9 [−1.3, +3.3]	+1.5 [+0.0, +3.2]	+2.1 [−0.4, +4.6]	−0.6 [−3.1, +2.0]	4.7	16.2	0.710	1
BFCL q8 AWQ-W4	788	4.1	2.7	6.7	+1.4 [−0.3, +3.3]	548	+2.0 [−0.4, +4.7]	+1.8 [+0.0, +3.7]	+3.6 [+0.9, +6.5]	+0.2 [−2.6, +3.1]	7.8	19.9	0.607	0
BFCL W4 pool					+0.1 [−1.0, +1.2]		+0.3 [−1.2, +1.8]	+0.9 [−0.2, +2.0]	+2.1 [+0.3, +3.9]	−0.6 [−2.3, +1.0]	10.3			
ALFWorld q8 GPTQ-W4	134	9.0	12.7	21.6	−3.7 [−11.2, +4.5]	115	−4.3 [−13.3, +5.0]	+7.0 [−2.6, +16.7]	+9.6 [−0.9, +19.7]	−11.3 [−20.9, −1.8]	40.0	46.4	0.137	4
ALFWorld q8 AWQ-W4	134	10.4	14.2	24.6	−3.7 [−11.9, +4.5]	119	−4.2 [−13.6, +4.9]	+5.0 [−2.6, +12.2]	+4.2 [−5.9, +13.7]	−9.2 [−18.8, +0.0]	32.8	42.9	0.237	3
ALFWorld W4 pool					−3.7 [−9.7, +1.9]		−4.3 [−11.1, +2.2]	+6.0 [−0.5, +12.6]	+6.9 [−2.1, +15.5]	−10.3 [−17.7, −2.9]	36.4			
BFCL q4 GPTQ-W3^{s}	591	6.1	3.6	9.6	+2.5 [+0.2, +5.1]	540	+2.8 [+0.2, +5.5]	+5.2 [+2.9, +7.7]	+8.9 [+5.7, +12.2]	−2.4 [−5.7, +0.7]	13.0	23.3	0.443	1
BFCL q8 GPTQ-W3^{s}	788	5.2	4.1	9.3	+1.1 [−1.0, +3.3]	696	+1.3 [−1.1, +3.7]	+1.4 [−0.1, +2.9]	+3.4 [+0.9, +6.0]	−0.1 [−2.6, +2.4]	8.3	20.4	0.592	1
BFCL q4 KV4^{s}	788	0.3	0.0	0.3	+0.3 [+0.0, +0.6]	787	+0.3 [+0.0, +0.6]	+2.9 [+1.9, +4.1]	+20.3 [+17.6, +23.0]	−2.7 [−3.9, −1.5]	20.5	20.9	0.020	0
BFCL q8 KV4^{s}	788	0.3	0.0	0.3	+0.3 [+0.0, +0.6]	788	+0.3 [+0.0, +0.6]	+1.8 [+1.0, +2.7]	+14.5 [+12.1, +17.0]	−1.5 [−2.5, −0.6]	14.7	15.2	0.029	0
ALFWorld q8 GPTQ-W3^{s}	134	6.7	3.0	9.7	+3.7 [−1.5, +9.7]	130	+3.8 [−1.5, +9.8]	+38.5 [+29.8, +46.9]	+42.3 [+33.6, +50.8]	−34.6 [−43.5, −25.4]	30.8	34.1	0.098	0
ALFWorld q8 KV4^{s}	134	0.0	0.0	0.0	+0.0 [+0.0, +0.0]	134	+0.0 [+0.0, +0.0]	+38.1 [+30.6, +45.5]	+53.0 [+44.8, +61.2]	−38.1 [−45.5, −30.6]	26.9	26.9	0.000	0

Table 8: Replacement and four-branch estimands on BFCL multi-turn and ALFWorld (no LLM user simulator; one run per item; 95% item-bootstrap intervals). All items: rescue r, damage d, turnover M (% of all items) and net G_{all} (pp), non-divergent items counting as zero. Divergent items (supplementary, exploratory, post hoc; pp, divergent items): Off-diag.: share with $\Delta_{\text{Q}} \neq \Delta_{\text{B}}$; exp. and κ: marginal-preserving pairing agreement control; rev.: strict reversals. Pools: equal weight per cell. ^{s}Stress and unrotated KV4 cells; in KV4 $Y_{\text{QQ}} = 0$ on every state, so the table reduces to H.

Cell	τ^2-bench retail (Bank P)	τ^2-bench airline (Bank P)	telecom dev (Bank P)	Bank U (retail / airline)	BFCL / ALFWorld
q4 GPTQ-W4	58 t · 727 st · 20/20 b	27 t · 375 st · 20/20 b	holdout only	3 b · 393 / 177 rows	788 items · 573 st / –
q4 AWQ-W4	58 t · 986 st · 20/20 b	27 t · 453 st · 20/20 b	–	3 b · 393 / 177 rows	788 items · 642 st / –
q4 KV8	58 t · 412 st · 22/22 b	27 t · 181 st · 22/22 b	–	3 b · 393 / 177 rows	– / –
q8 GPTQ-W4	58 t · 1017 st · 20/20 b	27 t · 456 st · 20/20 b	80 t · 477 st · 6/6 b; holdout	3 b · 393 / 177 rows	788 · 533 st / 134 · 115 st
q8 AWQ-W4	58 t · 1087 st · 20/20 b	27 t · 475 st · 20/20 b	80 t · 480 st · 6/6 b; holdout	3 b · 393 / 177 rows	788 · 548 st / 134 · 119 st
q8 KV8	58 t · 585 st · 22/22 b	27 t · 270 st · 22/22 b	–	3 b · 393 / 177 rows	– / –
q4 GPTQ-W3	58 t · 162 st · 3/3 b	27 t · 70 st · 3/3 b	–	–	591 · 540 st† / –
q4 KV4-rot	58 t · 164 st · 3/3 b	27 t · 69 st · 3/3 b	–	–	– / –
q8 GPTQ-W3	58 t · 166 st · 3/3 b	27 t · 76 st · 3/3 b	–	–	788 · 696 st / 134 · 130 st
q8 KV4-rot	58 t · 165 st · 3/3 b	27 t · 74 st · 3/3 b	–	–	– / –
q8 RTN-W3	58 t · 174 st · 3/3 b	27 t · 81 st · 3/3 b	–	–	– / –
q4 / q8 KV4 (unrotated)	–	–	–	–	788 · 787 / 788 · 788 st; q8 ALFWorld 134 · 134 st
q14 GPTQ-W4	–	–	–	–	788 · 471 st / –

Table 9: Completed crossed-intervention coverage (t = tasks, st = states, b = complete/found blocks; BFCL and ALFWorld: items · states, one run). Bank U: blocks completed and state rows per domain; blocks f01–f03 are analysed (Appendix J). "holdout": the cell is in the telecom holdout (Appendix N). Qwen3-32B has end-to-end runs only; Llama-3.1-8B stopped at its fork gate (Limitations); the q14 (Qwen3-14B) BFCL row is a within-family scale check. q4 RTN-W3 has no complete Bank P block and is omitted. The telecom development results (Appendix D.3) use the 6 blocks listed. †The BFCL base category of q4 GPTQ-W3 was dropped by the QQ no-op rule (Appendix G.3). The end-to-end q4 AWQ-W3 run is incomplete (two airline episodes hit 600 s serving timeouts), and so are the q4 RTN-W2 stress runs (context-window overflows of the degenerate model and a truncated result file); neither is reported.

ference, 0.46 for a flip given a difference and 0.55 for a rescue given a flip, the last on 3 to 39 flips per cell. Paired SpinQuant-W4A4 − QuaRot-W4A4 on 255 identical states (t over three block differences; crossed task × block bootstrap): KL −34 [−57, −11] (crossed [−56, −11]) $\times 10^{-3}$; difference rate +0.8 [−11.4, +13.0] ([−8.2, +9.4]) pp; flip rate given a difference +2.7 [−10.2, +15.3] ([−7.5, +11.4]); r_{cc} +0.4 [−3.0, +3.8] ([−4.3, +4.7]); d_{cc} +1.6 [−2.9, +6.0] ([−2.7, +5.5]); D_{cc} +1.2 [−3.9, +6.2] ([−4.7, +7.1]), equal to P2.

KIVI. Our implementation is a replay-compatible variant: at every decode step it reconstructs the quantized view that official KIVI's token-by-token decode would hold at that position from the full-precision cache, rather than persisting a compressed cache across turns, so every token, prompt or generated, attends with that view. The KIVI-2 contrast (P3) rests on two blocks: the third lost one arm to a server timeout that recurred on a single-request re-run, and both admitted blocks depend on the re-run of arms lost to server failures (strict rule: not estimable). Net of the kernel path, the quantization effect is +0.6 [−51.7, +52.9] (KIVI-2) and +2.4 [−5.4, +10.1] pp (KIVI-4), and the unquantized kernel path itself has $D_{cc} = -1.2$ [−6.2, +3.9].

Mistral population, scoring and coverage. The Mistral panel applies the same row rule to Mistral's own BF16 trajectories on the 85 retail and airline tasks. Seven BF16 source rows are missing because the source episode ended in an invalid tool call produced by the model itself (airline task 44 in all six blocks, airline task 13 in f01). A dated amendment, fixed from the BF16

source runs before any Mistral fork output existed, excludes both tasks in every block and cell, leaving 83 tasks. The selection was outcome-blind, but BF16 Mistral fails these tasks structurally, so the estimates describe this conditional population, not the 85-task panel, and are not compared with q8 as if the populations were equal. A branch that ended because the harness could not accept the agent's own message (an empty or invalid tool name, or an empty message) is scored as a failure, as in end-to-end evaluation. This rule was added after the Mistral phase-1 results had been read; without it the Mistral primary comparison is not estimable from phase 1 (one complete block). The analysis without the rule (which leaves such an arm missing and fails its block) and one applying the rule only to data not yet read when it was fixed are reported alongside; both admit the same three blocks for GPTQ − RTN ($+1.2$ [-13.8, $+16.2$]). Coverage: GPTQ-W4 has 498 states on six blocks and RTN-W4 415 states on five.

J Bank U per Sub-Bank

Table 10 gives the per-state turnover behind Appendix D.4, over blocks f01–f03. Exact action agreement is 0.43–0.83 per cell and sub-bank.

Cell	U1 (n=255)	U2B (n=255)	U2Q (n=60)
q4 GPTQ-W4	.090 (11/12)	.082 (10/11)	.083 (4/1)*
q4 AWQ-W4	.082 (7/14)	.078 (7/13)	.083 (3/2)
q4 KV8	.047 (4/8)	.039 (5/5)	.017 (0/1)
q8 GPTQ-W4	.047 (8/4)	.067 (9/8)	.050 (3/0)*
q8 AWQ-W4	.086 (14/8)	.090 (15/8)	.067 (3/1)
q8 KV8	.035 (5/4)	.031 (6/2)	0 (0/0)

Table 10: Bank U, blocks f01–f03: per-state turnover M (rescues/damages), task-equal (one state per task and block; 85/85/20 states per block). G lies within ± 0.028 in U1 and U2B and within ± 0.050 in U2Q, and every interval includes 0. Intervals are t intervals over the three block means and are wide (e.g. q4 GPTQ-W4 U1: .090 [.029, .151]). *GPTQ-W4's U2Q states are its own U1 states, with concordant outcomes on 20/20 duplicates per block; they are counted once in cross-sub-bank totals.

Text content. Over f01–f03, an above-median relative length difference between the two texts co-occurs with outcome change in U1 and U2B of both models (flipping pairs among above- vs. below-median pairs: q4 U1 30/154 vs. 15/155; q8 U1 22/101 vs. 6/101; q4 U2B 33/138 vs. 11/139; q8 U2B 16/99 vs. 9/100). In f01 alone this was visible only for q4. Question, confirmation-request and new-entity flags show no consistent association. The features are regular expressions, and this is descriptive.

Model:sub	(i) identical	(ii) same tool structure, text	(iii) different tool or args	(iv) act vs. talk
q4:U1	399: 0	309: 45 (.146)	19: 1 (.053)	38: 10 (.263)
q8:U1	483: 0	202: 28 (.139)	47: 11 (.234)	33: 4 (.121)
q4:U2B	430: 0	277: 44 (.159)	24: 4 (.167)	34: 3 (.088)
q8:U2B	470: 0	199: 25 (.126)	40: 9 (.225)	56: 14 (.250)
q4:U2Q	108: 0	63: 10 (.159)*	5: 0	4: 1 (.250)
q8:U2Q	129: 0	33: 6 (.182)*	11: 1 (.091)	7: 0
unique rows	1943: 0	1045: 150 (.144)	143: 26 (.182)	169: 32 (.189)

Table 11: Which kind of action difference changes the outcome at ordinary states (Bank U, blocks f01–f03). Entries: "state rows: outcome-changing rows (flip rate)", pooled over the three cells of a model within a sub-bank. The last row counts each (cell, state) once: q4/q8 GPTQ-W4's U2Q states repeat their U1 states (* includes 5 and 3 repeated class-(ii) flips), so 216 flipping rows become 208. In block f01 alone the corresponding counts were 39 of 56 flips in class (ii), with rates .126/.109/.200 for (ii)/(iii)/(iv); the 58 f01 flips scored in Appendix D.5 include the two duplicated U2Q rows.

Analysis	flips	retained	rev.	vanished (s/f/tie)	ctrl	new flips (resc./dam.)
All replies	39	7 [0.07, 0.34]	6	26 (15/2/9)	39	11 [0.15, 0.45] (7/4)
Flagged dropped	37	6 [0.06, 0.32]	6	25 (15/2/8)	39	11 [0.15, 0.45] (7/4)

Table 12: Bank U simulator control (block f01; Sonnet 5.5 relay user, $R = 3$). Fates use the sign of the replicate-mean flip indicator; s/f: both arms succeed/fail. "Flagged dropped": the two arm replicates with a mechanically flagged reply are set to missing. Replicates of one arm disagree in 16/39 (BQ) and 16/39 (QQ) flip states and in 10/39 and 8/39 control states. Per single replicate, retained flips are 5, 7 and 4 of 39.

Bank U simulator control: sensitivity and scope. Dropping two mechanically flagged replies leaves the reading of Appendix D.4 unchanged (6, 6 and 25 of 37; Table 12). In either direction, 13 of the 39 original flip states flip again, against 11 of 39 new flips at matched non-flip states; this does not show that the two groups are equivalent. The relay's responders may have seen other requests. The control's scope is narrow: fixed histories from the original simulator, a stochastic relay rather than API Sonnet or real users, block f01 only and three replicates.

Shared fragility across quantizers (a lead). On the 255 shared q4 U2B states of f01–f03, cell pairs co-flip 21 times, against about 3.3 expected under independence, and 16 of the 21 co-flips go in the

same direction. They come from 15 distinct states, 3 of which flip under all three cells. On q8 U2B, where every f01 flip was single-cell, 10 states now flip under two or more cells (14 pairwise co-flips against about 2.8 expected; 10 of 14 in the same direction). Joint flips almost never share an identical quantized action across cells (0 of 21 in q4, 2 of 14 in q8). We still read this as a lead toward *state-level* fragility, where different actions at the same state flip the outcome, and not yet as a finding: the co-flips are counted on one fixed set of states per block, with no correction.

A common controller (Q0). On the frozen 25% subsample of U2 in f01–f03, also continued by a common controller Q0, B's action leads to the identical outcome in every cell on all 146 shared states, which checks determinism. Turnover under Q0 is not consistently above or below the native turnover on the same rows (q4 AWQ-W4 U2B 8/58 vs. 5/58; q8 AWQ-W4 U2B 4/52 vs. 5/52; q4 KV8 U2B 1/58 vs. 2/58), so the f01 observation that Q0 turnover is at least the native one does not hold over three blocks. With 17–58 states per cell and sub-bank, this is too small to say whether the controller changes which actions matter.

K Concentration of Turnover in Tasks

This appendix gives the exploratory concentration analysis noted in Appendix D.3 (development data, 85 tasks $\times$ 20 blocks). The per-task dispersion of M is 4.0–8.7 times its binomial null in the W4 and KV8 cells, and the top 20% of tasks hold 57–71% of all flips. We split the 20 blocks into two disjoint halves of 10 nonces. Task-level M correlates across halves (Spearman 0.52–0.86). Tasks that were net-rescue in one half have a mean net gain in the other half that is 14–51 pp higher than that of net-damage tasks, in all 12 cell-directions. A Freedman–Lane test conditions on measured baselines: QQ success propensity and BF16 success, plus trajectory length in a post-hoc variant. With them, one half's task rate still predicts the other's in all 6 configurations $\times$ 2 directions for both M and G (Holm-adjusted). This is a *retrospective conditional association* on viewed development data. It is carried by retail. Airline alone gives 8 of 12 for M (5 after Holm), and 3 of 12 under a heteroscedasticity-robust wild bootstrap. Direction persistence beyond flip opportunity holds more often for rescues than for damages, in 11 versus 6 of the 12 cell-directions. Different quantizers flip the same tasks: the task-identity null is rejected in 7 of 8 quantizer pairs. When both flip, however, they agree in direction only 42–81% of the time. This overlaps with the stable failure set of Jang et al. (2026); unlike that work, we record the direction of each flip.

L Trigger Headroom and a KL Reference Row

Table 13 compares the frozen trigger of §6 (frozen rule, unchanged) with two references on the same 1020 f02/f03 U1 rows, which hold 38 rescues and 31 damages. The *matched action-replacement oracle* selects, per cell, exactly as many rows as the frozen rule did, taking only rows with $G > 0$ and filling with $G = 0$ rows when a cell has too few rescues; it never selects a damage. It is hindsight selection, not a policy. At the 10% and 20% budgets every rescue fits. The *three-option menu oracle* $E[\max(G, H, F, 0)]$ is the share of states where some arm beats QQ. On Bank P divergent states it is 0.202 over the four W4 cells (state-weighted, the one exception to the paper's equal-weight pools) against 0.117 for the action alone (1127 vs. 654 of 5576 states), and 0.191 vs. 0.113 over the six core cells. It is hindsight selection that also allows different replacement types, with different costs, so it is not a same-cost deployment comparison and not comparable with the action rule.

The *KL reference row* is a post-hoc auxiliary baseline. Its score is a truncated $\mathrm{KL}(\mathrm{B}\|\mathrm{Q})$ over the union of both models' top-20 next-token sets plus a tail bucket, per a_{Q} token, teacher-forced on identical prompt ids. It needs a BF16 prefill of the prompt and a_{Q} at every decision (here 6709.3 prompt and 82.3 action tokens on average), so it is not Q-only. It captures turnover above chance (33 vs. 16.7 at 20%) but no significant net gain (Holm $p = 1.0$). Three aggregations over the a_{Q} tokens were fit on f01 only, with per-cell thresholds and the frozen rule's selection criterion; the maximum was chosen, and all of f02/f03 is reported. The two candidates not chosen on f01 look better on f02/f03 (e.g. the mean at 20%: net $+9$, Holm $p = .0852$), but selecting them now would be selection on validation data. Because the f02/f03 outcomes had been seen, the row is post hoc, and it is not a DMR reproduction, which trains a router.

Budget	sel.	same per-cell selection counts: oracle r/d/net (pp)	frozen r/d/net (pp)	captured	sel.	KL reference (max. per-token KL; needs BF16; post hoc): net vs. random (p; Holm)	r/d	turnover vs. random (p)
5%	50	34/0/+34 (+3.33)	6/2/+4 (+0.39)	4/34	72	−1 vs. +0.45 (.8337; 1.0000)	5/6	11 vs. 4.58 (.0038)
10%	114	38/0/+38 (+3.73)	7/5/+2 (+0.20)	2/38	126	+1 vs. +0.82 (.5483; 1.0000)	8/7	15 vs. 7.54 (.0048)
20%	222	38/0/+38 (+3.73)	16/11/+5 (+0.49)	5/38	257	+3 vs. +1.59 (.3984; 1.0000)	18/15	33 vs. 16.73 (.0000)

Table 13: Frozen Q-only trigger against a matched action-replacement oracle (hindsight; same per-cell selection counts) and a post-hoc KL reference row, on f02/f03 U1 (1020 rows). Captured: frozen net over oracle net. Random: matched-count per-cell random-trigger expectation; p: exact one-sided hypergeometric, Holm over the three budgets.

Ctrl	Contrast	resc./dam.	turnover	net [95% CI]
B	INV vs VAL	4/3	7/30	+0.03 [−0.13, +0.20]
B	PP vs VAL	3/2	5/30	+0.03 [−0.10, +0.20]
B	INV vs PP	2/2	4/30	0.00 [−0.13, +0.13]
Q	INV vs VAL	2/2	4/30	0.00 [−0.13, +0.13]
Q	PP vs VAL	3/2	5/30	+0.03 [−0.10, +0.17]
Q	INV vs PP	2/3	5/30	−0.03 [−0.20, +0.10]
Controller dependence D ($\neq 0$ in 8/30 states)				+0.03 [−0.17, +0.27]

Table 14: Defect injection on 30 frozen q4 states (development, one block): per-state rescue/damage of one arm against another, under the BF16 (B) or quantized (Q) continuation controller. $D = [Y(\text{INV}, B) - Y(\text{VAL}, B)] - [Y(\text{INV}, Q) - Y(\text{VAL}, Q)]$. Task-clustered bootstrap intervals; all exact McNemar $p \geq 0.25$.

M Controlled Injection and Process Audit

A natural explanation of the two-sided turnover is a local-defect story: the quantized agent makes a local error, and the task fails unless the continuation recovers. Two analyses constrain this explanation without identifying a mechanism.

Controlled defect injection. We froze 30 ordinary states from q4 GPTQ-W4 trajectories, all from one development block. At each state we forced one of three actions as the next step: an INValid action, a matched VALid action, or a process-perturbation control (PP) that repeats the last successful read. Either B or Q then continued. The defect categories were wrong_id, wrong_tool and premature_write, with 10 states each. Every state passed a replay audit of 22 recorded fields (VAL×Q against the original episode). Three fields were not recorded: the counters right after the restore, the rendered-prompt hash sequence and the tool-schema hash sequence.

Result: two-sided turnover, no clear error-specific difference detected. Injected defects change the outcome in 7/30 states under B (4 rescues, 3 damages) and 4/30 under Q (2 and 2). The process-perturbation control changes 5/30 under both controllers (Table 14). The error-specific effect, INV minus PP, is 0.00 [−0.13, +0.13] under B and −0.03 [−0.20, +0.10] under Q. The controller dependence D is nonzero in 8 of 30 states, but its mean is +0.03 [−0.17, +0.27]. So injected defects cause heterogeneous two-sided turnover, and at $n = 30$ states we detect no clear difference from a generic process perturbation and no mean controller difference. The intervals are wide, so this does not show that errors and generic perturbations are equivalent.

Manipulation checks limit the categories. Two of 30 states failed a manipulation check: an injected write was rejected, and a valid reference itself returned "not found". Restricting to the 28 passing states leaves the reading unchanged. The wrong_tool category mostly produced argument errors: 8 of 10 injected calls failed with an unexpected-keyword error. We therefore call it a *tool-call/schema error perturbation*, not a wrong-tool test. premature_write is mostly failure→failure: the valid arm under Q already fails in 7/10 of these states, and turnover is 0/10 under both controllers. The event chain error → correction → related requirement satisfied completes in only 6 (B) and 7 (Q) of 29 injected errors, although 17 are corrected under each controller. Injections add tool errors (+0.43 under B, +0.57 under Q per branch).

Replication on q8 GPTQ-W4. The same design on 30 frozen q8 states (one development block; 10 per category) gives the same reading (Table 15). All 30 states pass the replay audit. Injected defects change the outcome in 5/30 states under each controller (3 rescues, 2 damages), the process-perturbation control in 6/30 (B) and 5/30 (Q). The error-specific effect is −0.03 [−0.13, +0.07] under B and −0.13 [−0.30, +0.03] under Q; the latter interval still allows a sizeable error-specific harm, and D is nonzero in 6 of 30 states with mean 0.00 [−0.17, +0.17]. Under Q, the

Ctrl	Contrast	resc./dam.	turnover	net [95% CI]
B	INV vs VAL	3/2	5/30	$+0.03$ $[-0.10, +0.17]$
B	PP vs VAL	4/2	6/30	$+0.07$ $[-0.07, +0.23]$
B	INV vs PP	1/2	3/30	-0.03 $[-0.13, +0.07]$
Q	INV vs VAL	3/2	5/30	$+0.03$ $[-0.10, +0.17]$
Q	PP vs VAL	5/0	5/30	$+0.17$ $[+0.03, +0.30]$
Q	INV vs PP	2/6	8/30	-0.13 $[-0.30, +0.03]$
Controller dependence D ($\neq 0$ in 6/30 states)				0.00 $[-0.17, +0.17]$

Table 15: Defect injection on 30 frozen q8 GPTQ-W4 states (development, one block), as in Table 14. Task-clustered bootstrap intervals; exact McNemar $p \geq 0.0625$.

process-perturbation control itself rescues 5 states and damages none ($+0.17$ [$+0.03$, $+0.30$], McNemar $p = 0.0625$). Manipulation checks fail more often than in q4: 7 of the 10 premature writes failed the check (6 were rejected by the environment, 1 executed without changing the database), and 2 valid references returned "not found". On the 21 passing states, INV vs VAL turnover is 5 (B) and 4 (Q), and $D \neq 0$ in 5. The q8 premature_write category therefore mostly did not inject a write.

Process audit (LLM-annotated). We sampled 80 development fork states, 20 per outcome sign. Two independent Sonnet 5.5 instances annotated them blind, a third instance adjudicated, and rule-based programmatic labels were added. These annotations are a *process description* conditional on outcome-defined groups. They are not causal evidence, not prevalences and not human labels. Three patterns stand out (Table 16). First, local defects and final failure do not correspond one-to-one. In damage states, where Q's action led to success and B's to failure, Q's first-action defect is corrected later in 10 of 20 QQ branches. Second, whether a write executes does not separate the failing from the succeeding continuation of the same state (rescue 17 vs. 17; damage 15 vs. 14). Third, the failing continuation more often carries the rubric label `unintended_or_unconsented_change` (rescue: QQ 9 vs. BQ 2, paired difference $+0.35$ [$+0.08$, $+0.57$]; damage: 1 vs. 9, -0.40 [-0.62, -0.11]). In damage states, 7 of these 20 BQ branches carry the label on writes *with* recorded consent. That is the semantic "wrong change" part of the label, and it overlaps with the database-defined outcome by construction. Loops and budget exhaustion also mark failing branches (rescue: QQ 5 vs. BQ 0). We make no claim about consent. The programmatic consent field agrees poorly with the annotators, so we use it only as a mechanical row. Re-labelling 30 states with fresh instances of the same model family reproduces the key pattern.

M.1 Process audit table

The core "local defect" combines groundedness, argument validity and permission/consent judgements. The tool-validity field (inter-annotator $\kappa = 0.29$) is excluded from it. Mechanical fields agree with the programmatic verifier (write executed vs. the rubric's irreversible-change field, $\kappa = 0.99$). The consent fields agree only moderately ($\kappa = 0.44$ on writes both see).

N Holdout Pre-Registration

The protocol was frozen before any holdout unit was run, and its hash was recorded in the split manifest. It fixes the following design:

- tasks, cells, weights and nonce blocks;
- two fixed halves of six blocks each;
- an outcome-blind repair rule, with reserve blocks used only for infrastructure repair;
- the decisional crossed interval (two-way ANOVA moments, normal 90%);
- the readings: positive if $L > 0$, above 3 pp if $L > 3$, small if the interval lies inside ± 3 pp, otherwise inconclusive;
- the secondary families S1 (both $r > 0$ and $d > 0$) and S2 (persona-stratified permutation, 10,000 draws, no outcome screen);
- six stratum families, with Holm correction within each.

The analysis script refuses holdout data unless the hash matches. Two task-cell pairs were excluded as structural, and the complete-case analysis is identical.

Replay acceptance on the holdout (post hoc audit). Y_{QQ} is the one arm taken from the original episode rather than re-executed, and an exploratory count showed only-QQ successes more often than random labelling would give in two q8 holdout cells. We therefore audited the holdout's acceptance records after the analysis, with a script that outputs only consistency checks and counts. The fixed hash rule selected 135 of the 1202 divergent states for the QQ no-op; all 135

Label (source)	Comparison	rescue	damage	both succeed	both fail
`unintended_or_unconsented_change` (LLM)	QQ vs BQ	9 vs 2, $+0.35$ [$+0.08$, $+0.57$]	1 vs 9, -0.40 [-0.62, -0.11]	1 vs 1	5 vs 7, -0.10 [-0.35, $+0.17$]
write executed (programmatic)	QQ vs BQ	17 vs 17, 0.00 [-0.25, $+0.25$]	15 vs 14, $+0.05$ [-0.26, $+0.35$]	10 vs 9	11 vs 13
executed write w/o recorded consent (prog.)	QQ vs BQ	9 vs 5, $+0.20$ [-0.06, $+0.43$]	3 vs 3, 0.00 [-0.21, $+0.21$]	3 vs 2	4 vs 3
loop or budget exhaustion (LLM)	QQ vs BQ	5 vs 0, $+0.25$ [$+0.04$, $+0.47$]	0 vs 2, -0.10 [-0.30, $+0.08$]	1 vs 2	7 vs 4
action judged indefensible (LLM)	Q vs B action	7 vs 8, -0.05 [-0.37, $+0.28$]	12 vs 5, $+0.35$ [-0.05, $+0.65$]	10 vs 6	12 vs 9
local defect, core (LLM)	Q vs B action	4 vs 6, -0.10 [-0.33, $+0.14$]	12 vs 7, $+0.25$ [-0.09, $+0.53$]	6 vs 3	11 vs 7
QQ first-action defect corrected later (LLM)	single branch	4/20 [0.08, 0.42]	10/20 [0.30, 0.70]	4/20	7/20

Table 16: Process audit of 80 development fork states, 20 per outcome sign (rescue: $Y_{\mathrm{BQ}} = 1, Y_{\mathrm{QQ}} = 0$; damage: $Y_{\mathrm{BQ}} = 0, Y_{\mathrm{QQ}} = 1$). **LLM-annotated** by Sonnet 5.5 (two independent blind instances and an adjudicating instance), plus rule-based programmatic labels; no human labels. Counts are first vs. second of $n = 20$ states. Intervals are within-state paired differences with Newcombe 95% CIs; single-branch rows use Wilson 95% intervals. This is a process description conditional on outcome-defined groups: not causal, not prevalence. The label name is the rubric category; we make no claim about consent.

were run and passed, none was missing, failed or non-computable, and no other state was re-run. Beyond the frozen checks (suffix, prompts, termination, outcome, budget counters), the 135 also match on reward, database state, the official reward, the final environment hash against a strict replay of the original episode, and the history prefix. The original QQ and the re-run arms are scored by the same official reward rule: recomputing it from the stored reward information reproduces the stored value for every arm. The audit read the same input files as the frozen analysis, so no repair under the protocol's missing-data rules was needed, and the pre-registered result stands as reported. The audit does not explain the only-QQ pattern; a real upgrade that damages the original policy would also produce it. Table 17 gives the confirmatory result. Table 18 gives all six stratum families (Holm within family), which are descriptive. Every level estimate is ≤ 0, and the Holm-significant levels are negative: step-budget T2 and T3, mobile-data issues (-9.09 pp) and the fewest-faults tercile (-6.37 pp).

Cell (weight)	units	excl.	G	r	d	ρ_c
q4 GPTQ-W4 (1/2)	396	1	-3.03 [-7.86, $+1.80$]	$+5.05$	$+8.08$	0.668
q8 GPTQ-W4 (1/4)	408	0	-5.64 [-9.33, -1.94]	$+4.41$	$+10.05$	0.374
q8 AWQ-W4 (1/4)	396	1	-6.57 [-11.98, -1.15]	$+6.31$	$+12.88$	0.644
Weighted (G_{all})			-4.57 [-7.17, -1.96]	$+5.21$	$+9.77$	0.588

Table 17: Telecom holdout, pre-registered analysis (34 tasks $\times$ 12 blocks; pp; crossed 90% intervals). Per-cell rows are descriptive; the weighted row is decisional (G_{all}, reading: inconclusive) and S1/S2 (both pass). Excl.: structurally excluded tasks (decision/trajectory count mismatch at search).

Post-hoc four-branch estimands on the holdout. Table 19 is a supplementary, descriptive output computed after the confirmatory analysis, with the same admission and units. It is not part of the pre-registered confirmation and changes neither the primary endpoint nor its reading. Its G is a ratio over divergent states, whereas G_{all} in Table 17 averages over all admitted units with non-divergent units counted as 0.

Family	Level	n	G [crossed 90%]	p	Holm p
step budget[a]	T1	138/129/110	-0.32 [-1.57, $+0.94$]	.6771	.6771
	T2	124/126/162	-2.30 [-3.88, -0.72]	.0165	.0495
	T3	123/144/124	-1.94 [-3.35, -0.54]	.0229	.0495
persona	Hard	11	-5.99 [-12.23, $+0.26$]	.1147	.3355
	None	12	-3.65 [-7.42, $+0.13$]	.1118	.3355
	Easy	11	-4.36 [-9.19, $+0.48$]	.1385	.3355
BF16 difficulty	0	20	-2.35 [-4.09, -0.62]	.0257	.0772
	low	11	-4.54 [-8.90, -0.19]	.0860	.1720
	high	3	-18.75 [-38.00, $+0.50$]	.1091	.1720
task length	T1	9	-5.09 [-9.81, -0.38]	.0755	.1511
	T2	8	-0.78 [-4.46, $+2.90$]	.7270	.7270
	T3	17	-6.17 [-10.49, -1.84]	.0189	.0567
issue type	mms	14	-1.15 [-3.33, $+1.04$]	.3880	.4434
	mobile data	11	-9.09 [-14.98, -3.20]	.0111	.0333
	service	9	-4.40 [-10.32, $+1.52$]	.2217	.4434
faults	T1	19	-6.37 [-9.91, -2.83]	.0030	.0091
	T2	10	-1.94 [-7.58, $+3.69$]	.5704	.5704
	T3	5	-2.92 [-5.18, -0.66]	.0337	.0674

Table 18: Telecom holdout strata: the six pre-registered families (descriptive; Holm within each family over its three levels). pp. For task-level families, n is the number of tasks and G is G_{all} over the tasks in the level. [a]Step budget (remaining steps at the divergence) is unit-level: n is the number of divergent units per cell (q4 GPTQ-W4/q8 GPTQ-W4/q8 AWQ-W4), and G is the level's contribution to G_{all}, so the three levels sum to G_{all}. Tertile cut points were fixed on telecom development tasks.

O Withdrawn, Negative and Supplementary Results

W$\times$KV interaction (not supported). A first q4 analysis found a W4$\times$KV4 interaction of -14.9 [-23.8, -6.0] pp ($k = 3$). Its KV16 reference lacked the Q/K rotation used for KV4, so rotation and quantization were confounded. With a matched rotated reference, the q4 interaction is -9.0 [-30.6, $+12.5$]. The pre-registered q8 replication gives $+2.1$ [-5.8, $+10.1$] over 6 blocks,

Cell (weight)	states	G	H	F	$G - H$	mean I	off-diag. %	exp. %	κ	rev. % (n)
q4 GPTQ-W4 (1/2)	385	-3.1 [-9.0, $+2.8$]	$+0.5$ [-5.4, $+6.4$]	$+1.8$ [-4.8, $+8.4$]	-3.6 [-9.1, $+1.8$]	-4.4 [-12.5, $+3.7$]	19.7	27.0	0.269	3.4 (13)
q8 GPTQ-W4 (1/4)	399	-5.8 [-10.2, -1.3]	-4.0 [-10.2, $+2.2$]	-5.3 [-11.6, $+1.1$]	-1.8 [-6.4, $+2.9$]	-4.5 [-10.4, $+1.4$]	19.5	23.1	0.155	3.0 (12)
q8 AWQ-W4 (1/4)	396	-6.6 [-13.0, -0.1]	-8.3 [-15.3, -1.3]	-8.3 [-15.5, -1.2]	$+1.8$ [-3.3, $+6.8$]	-6.6 [-14.5, $+1.4$]	23.7	26.0	0.088	1.0 (4)
Weighted pool	1180	-4.6 [-7.8, -1.5]	-2.8 [-6.3, $+0.6$]	-2.5 [-6.2, $+1.3$]	-1.8 [-4.8, $+1.1$]	-5.0 [-9.6, -0.4]	20.7	25.7	0.195	2.7

Table 19: Telecom holdout, *post-hoc* four-branch estimands on divergent states (descriptive; not part of the pre-registered confirmation). pp; crossed task×block 95% intervals; pool weights as pre-registered. Exp. and κ: marginal-preserving pairing agreement control; rev.: strict reversals.

crossed [-8.2, $+12.4$]. The replication is not supported, and we make no W×KV claim.

Static precision restoration (appendix only). We tested whether restoring selected modules to higher precision recovers a damaged RTN-W3 model. We report this line as a mechanism stress test, not a method. In RTN-W3 the gains largely tracked the number of restored bits. A random MLP arm also recovered much of the gap, and no arm beat whole-model GPTQ-W4 at matched storage. Its comparison with byte-matched random modules, incomplete when the line was paused, is reported below for the cells that have since finished.

Success equivalence. Not established (Table 1).

Supplementary perturbation analyses (exploratory). These analyses perturb the quantized weights directly instead of forking trajectories, and they enter no claim in the main text. Each derived checkpoint takes every decoder linear from the BF16 model, from Q, or from an interpolation between them; all other tensors come from the BF16 model, and the endpoints reproduce BF16 and Q bit for bit. Only finished end-to-end runs enter: q4 and q8, 85 tasks on each of three nonce blocks per cell (q3–q5 for W3, the development blocks f01–f03 for W4), with outage episodes excluded and context overflow counted as failure. Intervals are fixed-pool t intervals over the three blocks, with crossed task×block intervals alongside; there is no correction for multiplicity.

Restoration location versus random modules. Restoring the BF16 weights of one layer third, of attention only or of the MLP only is compared with restoring randomly chosen modules with the same number of BF16 parameter bytes (mean over the finished random draws, up to three), on the same task×block units (Table 20). In RTN-W3, where Q succeeds in 12.9% (q4) and 13.3% (q8) of episodes against 35.7% and 30.6% for BF16, restoring attention beat byte-matched random modules by $+9.8$ [$+6.1$, $+13.6$] pp (q4 and q8 pooled; crossed [$+4.5$, $+15.1$]), and restoring the last third did worse than random, -8.6 [-13.4, -3.8] (crossed [-12.9, -4.7]); the middle third ($+4.5$, crossed [-0.1, $+9.4$]) and the MLP (-5.4 [-14.1, $+3.3$]) are not resolved. In GPTQ-W3, where Q and BF16 succeed about equally often, and in q4 GPTQ-W4 and q8 AWQ-W4, none of the 30 contrasts excludes zero under both intervals; of all 39, the three that do are RTN-W3 contrasts. At the first divergent state itself (one decision, one random draw), restoring attention returned Q's action to B's more often than random modules did (q4 RTN-W3 $+20.7$ [$+6.3$, $+35.0$] pp; GPTQ-W3 pooled $+35.7$ [$+27.3$, $+44.1$]), and restoring the last third did so less often (q4 RTN-W3 -44.6); in GPTQ-W3 this action effect has no counterpart in success ($+1.6$ [-4.8, $+7.9$]). Not finished, hence not reported: the q4 RTN-W3 early third, the third random draw for attention in both RTN-W3 models and for the q4 RTN-W3 MLP, and the q8 RTN-W3 pivot arms, so the pre-declared pivot-arm selection never ran.

Error dose. The residual intervention $W_\alpha = W + \alpha(W_\mathrm{Q} - W)$ scales the quantizer's own weight error, with $\alpha \in \{0, 0.25, 0.5, 1, 2\}$; it is not a bit width (Table 21). Agreement with BF16's actions fell at every step of α in all six cells, with both intervals of every step below zero; in q8 RTN-W3, the share of BF16's decisions before the first differing action is 1, 0.32, 0.16 and 0.04 at $\alpha = 0$, 0.25, 0.5 and 1. Success did not follow. No success step up to $\alpha = 0.5$ is resolved under both intervals (in RTN-W3 the point estimates at $\alpha = 0.25$ lie above BF16, 40.0 and 36.5% against 35.7 and 30.6%), and success fell between $\alpha = 0.5$ and 1 only in RTN-W3, by -20.8 [-22.5, -19.1] pp in q4 (crossed [-29.8, -11.8]) and -14.1 [-26.9, -1.4] in q8 (crossed [-24.7, -3.9]). In GPTQ-W3, q4 GPTQ-W4 and q8 AWQ-W4 no success step is resolved under both intervals; doubling the q8 AWQ-W4 error ($\alpha = 2$) changed success by -10.6 [-16.4, -4.7] pp (crossed [-20.8, 0.0]).

	RTN-W3 (q3–q5)			GPTQ-W3 (q3–q5)		q4 GPTQ-W4	q8 AWQ-W4
Located arm	q4	q8	q4+q8	q4	q8	(f01–f03)	(f01–f03)
Success Q / BF16 (%)	12.9 / 35.7	13.3 / 30.6		36.1 / 35.7	26.7 / 30.6	43.9 / 33.7	32.5 / 31.4
Early third	n.f.	−2.1 [−10.0, +5.8]		+2.7 [−3.3, +8.8]	+0.4 [−1.6, +2.3]	+5.6 [−1.7, +12.9]	−1.4 [−10.2, +7.3]
Middle third	**+6.9** [+0.1, +13.6]	+2.2 [−1.7, +6.2]	+4.5 [+2.7, +6.3]	−4.7 [−12.4, +3.0]	−4.3 [−16.8, +8.1]	−1.0 [−7.0, +4.9]	+1.3 [−5.3, +7.9]
Last third	−7.7 [−15.6, +0.2]	**−9.5** [−13.5, −5.6]	**−8.6** [−13.4, −3.8]	+1.6 [−8.7, +11.8]	−0.4 [−17.6, +16.8]	−0.3 [−10.8, +10.2]	−1.0 [−5.0, +2.9]
Attention	**+12.2*** [+4.1, +20.2]	+7.5* [−1.5, +16.4]	**+9.8** [+6.1, +13.6]	+5.9 [+0.8, +10.9]	−2.7 [−10.4, +4.9]	+0.8 [−18.4, +20.0]	−0.5 [−5.1, +4.1]
MLP	−9.6* [−21.8, +2.6]	−1.2 [−6.3, +4.0]	−5.4 [−14.1, +3.3]	−1.4 [−10.5, +7.6]	−3.0 [−4.1, −1.9]	+0.1 [−5.4, +5.7]	+6.0 [−7.2, +19.2]

Table 20: Restoration location versus byte-matched random modules (exploratory). Closed-loop success of the located arm minus the mean success of random module sets with the same BF16 parameter bytes (three draws; *two draws, the third did not finish), pp, paired on 253–255 task×block units per cell; fixed-pool 95% t interval over three blocks. Bold: the crossed task×block interval also excludes zero (crossed intervals and per-draw success in the analysis output). q4+q8: equal-weight pool. n.f.: not finished. GPTQ-W3 arms swap BF16 modules into the sequentially calibrated checkpoint (re-calibrated variants in the analysis output). In the W4 cells Q is not worse than BF16 on these blocks, so a difference there is not a recovery.

The $\alpha = 2$ arms of the four W3 cells did not finish. The action response to the error dose is thus monotone, whereas success shows only a drop between half and full RTN-W3 error.

Cell	$\alpha = 0$ (BF16)	0.25	0.5	1 (Q)	2
q4 RTN-W3	35.7 (1)	40.0 (0.362)	33.7 (0.225)	12.9 (0.088)	n.f.
q8 RTN-W3	30.6 (1)	36.5 (0.319)	27.5 (0.162)	13.3 (0.043)	n.f.
q4 GPTQ-W3	35.7 (1)	37.6 (0.508)	36.1 (0.406)	36.1 (0.234)	n.f.
q8 GPTQ-W3	30.6 (1)	29.4 (0.397)	29.0 (0.297)	26.7 (0.176)	n.f.
q4 GPTQ-W4	33.7 (1)	38.0 (0.572)	38.4 (0.504)	43.9 (0.424)	37.6 (0.238)
q8 AWQ-W4	31.4 (1)	31.0 (0.392)	32.5 (0.309)	32.5 (0.194)	22.0 (0.058)

Table 21: Error dose (exploratory): closed-loop success (%) of W_α and, in parentheses, action agreement with BF16 (share of BF16's decisions before the first differing action; 1 = identical). 255 task×block units per cell over three blocks; action agreement on 245 units for q4 RTN-W3, whose other 10 units contain an episode that ended in context overflow and has no message record. n.f.: not finished. Intervals of every step are in the analysis output.

Low-bit failure modes. Table 22 assigns every finished, non-outage retail and airline episode to one ending (descriptive). BF16 failures nearly all end normally in a wrong end state (57.6–63.9% of episodes; at most 6.3% end otherwise), and GPTQ-W3 keeps this profile (1.8–9.0% abnormal endings). RTN-W3 and q4 AWQ-W3 move failures to episodes that never end normally or never issue a parsed tool call. q4 RTN-W3 runs away (context overflow or the 200-step cap) in 32.9% of episodes; in 44.7% it ends normally after writing its tool calls as plain JSON, which the frozen tool parser does not accept. No q4 RTN-W3 episode issues a parsed write call, and none of its 33 successes is on a task that requires a write. q8 RTN-W3 runs away in 20.4% of episodes and hits the cap of 10 failed tool calls in 16.9%. In the W2 panel (24 tasks, one block, loop guard on), 66.7–87.5% of episodes run away. No episode ended on an unparseable output. KV-INT3 (q4, q8, q14), Gaussian weight noise on q14, q32 RTN-W3 and the q4 RTN-W2 panel had not finished.

Cell	n	Succ.	Run.	Err.	Text	No wr.	Tool	Args
q4 BF16	255	35.7	0.4	0.0	0.4	14.9	23.9	24.7
q4 GPTQ-W3	255	36.1	1.2	1.2	0.4	14.9	24.7	21.6
q4 AWQ-W3	253	19.4	16.6	0.0	30.8	32.8	0.4	0.0
q4 RTN-W3	255	12.9	32.9	0.0	44.7	9.4	0.0	0.0
q8 BF16	255	30.6	3.5	2.7	0.0	8.2	31.0	23.9
q8 GPTQ-W3	255	26.7	4.3	4.7	1.6	9.8	30.6	22.4
q8 AWQ-W3	255	14.9	3.1	11.4	0.4	13.3	35.3	21.6
q8 RTN-W3	255	13.3	20.4	16.9	6.7	12.2	16.1	14.5
q8 GPTQ-W2†	24	12.5	66.7	0.0	0.0	20.8	0.0	0.0
q8 RTN-W2†	24	0.0	87.5	0.0	0.0	12.5	0.0	0.0
q14 BF16	85	42.4	0.0	0.0	0.0	7.1	30.6	20.0
q14 GPTQ-W3	425	33.9	0.7	1.9	0.2	16.2	27.1	20.0
q14 RTN-W3	425	23.5	11.3	1.4	1.9	15.1	24.5	22.4
q32 BF16	85	38.8	0.0	0.0	0.0	10.6	31.8	18.8
q32 GPTQ-W3	170	40.0	0.6	1.2	0.0	4.1	28.2	25.9
q32 noise‡	82	17.1	12.2	9.8	9.8	24.4	11.0	15.9

Table 22: Low-bit failure modes (descriptive): share (%) of n finished, non-outage retail and airline episodes by how the episode ends, one category per episode, assigned in column order. Succ.: DB success. Run.: no normal ending (context overflow, 200-step cap, or the W2 loop-guard stop). Err.: stopped after 10 failed tool calls. Text: normal ending; a required write was never issued as a parsed call while the agent wrote tool calls as plain text. No wr.: normal ending; a write was required and none was attempted. Tool: the set of write tools called differs from the required set. Args: same write tools, wrong end state. q4 and q8: blocks q3–q5; q14 W3: five blocks; q32 GPTQ-W3: two blocks; q14 and q32 BF16: one run. †24-task W2 panel, one block, loop guard on. ‡Gaussian weight noise at three times the RTN-W4 error norm, one seed. Outage episodes excluded: 2 (q4 AWQ-W3) and 3 (q32 noise). A6000 runs only.

P Related Work Details

Flat Score, Amplified Failures (Jang et al., 2026) reports a two-sided paired task split (33 tasks

worse, 26 better for Gemma-4-31B on telecom) and recovery equally frequent at both precisions (55% after a failed call). *The Replay Gap* (Gonuguntla, 2026) forks live SWE-bench Verified trajectories at 30% and 70% of their length, re-executes the prefix in a fresh container, continues each fork with the other model of a pair (Qwen3-4B FP8 and Qwen3-14B AWQ), and compares against same-model control forks; swaps rewrite most post-fork actions, all five observed outcome flips occur in swap arms and none in 359 control forks, and log-stitching replay mispredicts outcomes.

Q Provenance

Every number in the paper is traceable to released analysis outputs. Figures are drawn by a released plotting script from those outputs, with assertions on every value the text quotes, and the TikZ figures are typeset from numbers in the LaTeX source.